EVOLUTIONARY COMPUTATION

LABORATORY 04

Candidate moves

for the Traveling Salesperson Problem with Node Costs

ASSIGNMENT 4 - REPORT

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-04-candidate-moves

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OBJECTIVE

Accelerate steepest 2-opt local search

Evaluate only moves introducing candidate edges.

The supplied instances contain 200 nodes, so every solution selects 100.

INSTANCES

TSPA and TSPB

200 nodes each - 100 selected

EXPERIMENT

k = 1, 5, 10, 15, 20, 30, 50

plus full-neighborhood baseline

Poznań University of Technology - Artificial Intelligence

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1. Problem description

The input consists of n nodes, each with integer planar coordinates (x, y) and an integer node cost.

The task is to choose exactly k = ceil(n / 2) distinct nodes and specify their cyclic visiting order.

The cycle must return from its last selected node to its first selected node.

OBJECTIVE

f(S) = cycle length(S) + sum of costs of selected nodes

DISTANCE

d(i,j) = round(sqrt((xi-xj)^2 + (yi-yj)^2))

Distances are calculated once
and stored in a distance matrix.

DATA SEPARATION

Optimization methods access
the distance matrix and node
costs, not the original coordinates.

For the supplied TSPA and TSPB instances, n = 200 and k = 100.

Lab 04 starts from random feasible cycles and compares full steepest search with a candidate-move version. The candidate version keeps only moves that introduce at least one candidate edge.

Solution representation

A solution is an ordered list of k distinct node indices.

The closing edge from the last node to the first is included in the objective.

All reported indices are zero-based, matching the supplied solution checker.

2. Candidate-move algorithm

Candidate-move steepest local search

The candidate method builds only eligible 2-opt and selected-to-unselected moves. The full-neighborhood baseline uses the same steepest loop but replaces CANDIDATE_NEIGHBORHOOD(S) with every valid move.

```
INPUT:
  D[u][v] rounded distance between vertices u and v
  C[u] cost of vertex u
  n number of vertices; k = ceil(n / 2)
  K candidate-list size (1, 5, 10, 15, 20, 30, or 50)
  S0 random feasible cycle with k distinct vertices
OUTPUT: locally optimal closed cycle S and objective F(S)

OBJECTIVE AND INDEX FUNCTIONS:
  F(S) = sum D[S[p]][S[next(p)]] + sum C[S[p]]
  next(p) = (p + 1) mod k; previous(p) = (p - 1 + k) mod k
  position[u] is the position of selected vertex u in S
  all routes below omit the repeated closing vertex S[0]

BUILD CANDIDATE EDGE SETS:
  for u in range(0, n):
    rank every v != u by D[u][v] # intra
    for v in the first min(K, n - 1) ranked vertices:
      insert undirected edge {u, v} into IntraCandidates
    rank every v != u by D[u][v] + C[u] + C[v] # inter
    for v in the first min(K, n - 1) ranked vertices:
      insert undirected edge {u, v} into InterCandidates
  Candidate edges are stored in IntraCandidates and InterCandidates.

DELTA FUNCTIONS (no neighboring route is reconstructed):
  delta_2opt(S, i, j) = D[S[previous(i)]][S[j]]
                      + D[S[i]][S[next(j)]]
                      - D[S[previous(i)]][S[i]]
                      - D[S[j]][S[next(j)]]
  delta_replace(S, p, v) = D[S[previous(p)]][v] + D[v][S[next(p)]]
                         + C[v] - D[S[previous(p)]][S[p]]
                         - D[S[p]][S[next(p)]] - C[S[p]]

CANDIDATE_NEIGHBORHOOD(S):
  selected[u] <- false for every u; position[u] <- UNDEFINED for every u
  for p in range(0, k):
    selected[S[p]] <- true; position[S[p]] <- p
  M <- empty list of moves; seen_2opt <- empty set
  for first in range(0, k):
    before <- S[previous(first)]
    for v in IntraCandidates[before]:
      if selected[v] == true:
        second <- position[v]
        if second > first + 1 and move is not wrap-around:
          add 2-opt move (first, second) if not in seen_2opt
    for v in IntraCandidates[S[first]]:
      if selected[v] == true:
        second <- previous(position[v])
        if second > first + 1 and move is not wrap-around:
          add 2-opt move (first, second) if not in seen_2opt
  for p in range(0, k):
    before <- S[previous(p)]; after <- S[next(p)]
    for v in InterCandidates[before] UNION InterCandidates[after]:
      if selected[v] == false:
        add replacement move (p, v) to M
  return M

CANDIDATE_STEEPEST_SEARCH(S0):
  S <- S0; current_cost <- F(S)
  while True:
    M <- CANDIDATE_NEIGHBORHOOD(S)
    best_delta <- 0; best_move <- NONE
    for each move m in M: # only candidates
      delta <- delta_2opt(S,m) or delta_replace(S,m)
      if delta < best_delta:
        best_delta <- delta; best_move <- m
    if best_move is NONE:
      break
    apply best_move to S; current_cost <- current_cost + best_delta
  append S[0] to close the cycle and return S, current_cost
```

3. Algorithm comparison

Objective score: average (minimum - maximum); lower values are better.

Algorithm	TSPA	TSPB
Candidate moves (k = 1, nearest)	119836.08(108540 - 138085) !	81385.45(72055 - 90620) !
Candidate moves (k = 5)	79905.04(74889 - 84157)	54893.31(52515 - 58105)
Candidate moves (k = 10)	77589.12(75168 - 81693)	53573.29(51374 - 56520)
Candidate moves (k = 15)	77384.93(74831 - 81611)	53472.66(51171 - 56408)
Candidate moves (k = 20)	**77336.46(74434 - 80540)**	**53461.22(51371 - 56469)**
Candidate moves (k = 30)	77355.34(74935 - 80342)	53515.31(51385 - 56469)
Candidate moves (k = 50)	77353.41(74935 - 80342)	53506.79(51385 - 56469)
Full neighborhood (k = infinity)	77354.36(74935 - 80342)	53505.13(51385 - 56469)

Runtime: average (minimum - maximum) milliseconds; lower values are better.

Algorithm	TSPA	TSPB
Candidate moves (k = 1, nearest)	**5.93(3.96 - 30.61) ms**	**5.37(4.09 - 7.79) ms**
Candidate moves (k = 5)	27.42(22.80 - 41.21) ms	25.10(19.05 - 39.89) ms
Candidate moves (k = 10)	52.49(41.16 - 101.95) ms	46.35(37.35 - 99.06) ms
Candidate moves (k = 15)	73.65(54.79 - 110.13) ms	65.64(50.07 - 96.86) ms
Candidate moves (k = 20)	93.56(76.23 - 131.95) ms	84.62(67.94 - 182.77) ms
Candidate moves (k = 30)	136.80(110.79 - 194.05) ms	118.16(90.92 - 210.63) ms
Candidate moves (k = 50)	198.19(161.17 - 268.11) ms	181.67(144.40 - 283.81) ms
Full neighborhood (k = infinity)	256.70(207.59 - 322.33) ms !	262.19(211.10 - 339.42) ms !

lowest average objective lowest average runtime lowest minimum score/time highest maximum score/time

Best results in the comparison

TSPA score: Candidate moves (k = 20) (77336.46).

TSPB score: Candidate moves (k = 20) (53461.22).

TSPA time: Candidate moves (k = 1, nearest) (5.93 ms).

TSPB time: Candidate moves (k = 1, nearest) (5.37 ms).

4. Best solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 1)

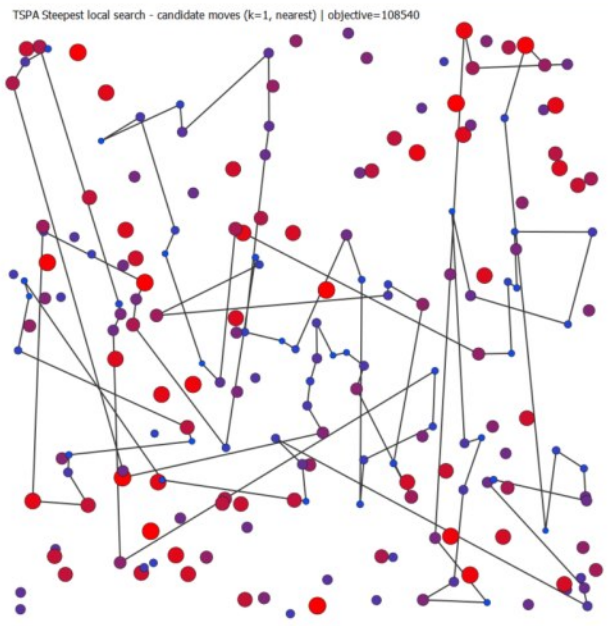
min=108540 max=138085 avg=119836.08

best = 108540

TIME (ms)

avg = 5.93

min = 3.96 max = 30.61



Best solution nodes:

```
157 71 115 112 139 35 67 144 143 33 58 84 176 174
92 160 83 123 7 48 31 77 161 125 41 140 53 179 52
177 36 131 99 65 134 194 8 9 54 162 16 51 87 148
90 68 60 82 188 120 127 97 141 145 168 63 149 187
135 73 45 78 153 193 42 89 170 146 159 130 107 178
29 17 55 24 197 138 57 111 182 151 106 150 164 85
47 147 1 50 3 93 81 196 28 180 132 76 91 80 157
```

Candidate moves (k = 5)

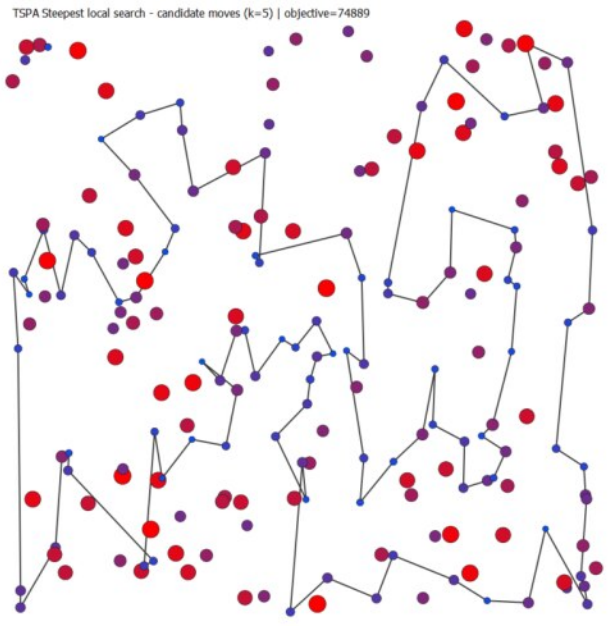
min=74889 max=84157 avg=79905.04

best = 74889

TIME (ms)

avg = 27.42

min = 22.80 max = 41.21



Best solution nodes:

```
164 1 169 3 50 93 43 106 113 89 42 151 150 176 174
72 134 58 84 36 131 33 143 51 16 139 4 71 157 91
80 38 76 132 90 123 152 108 49 60 149 63 116 168
129 104 69 95 97 127 2 171 30 187 141 87 102 180
28 156 35 185 67 144 52 177 179 53 140 45 135 73
114 181 192 158 182 57 64 107 78 193 153 170 146
159 130 47 37 147 55 29 101 178 136 186 9 54 8 142
164
```

Candidate moves (k = 10)

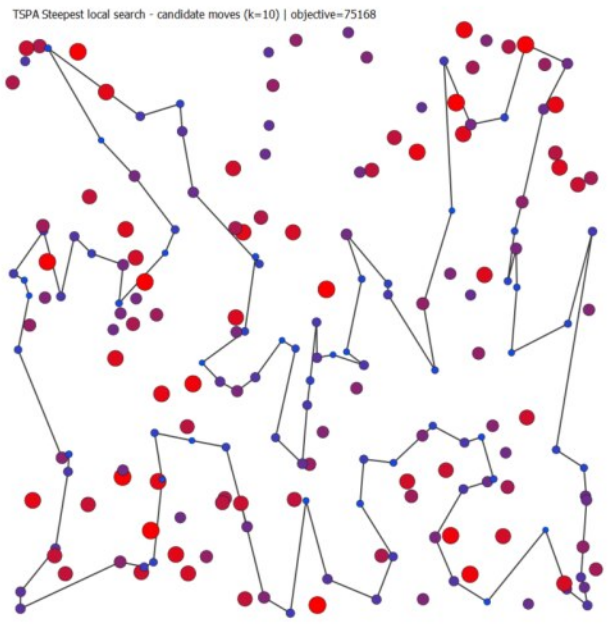
min=75168 max=81693 avg=77589.12

best = 75168

TIME (ms)

avg = 52.49

min = 41.16 max = 101.95



Best solution nodes:

```
182 138 151 42 89 106 150 176 72 134 84 58 158 192
181 135 114 25 70 98 87 141 30 187 2 171 92 104
129 69 127 97 95 168 63 149 116 60 49 108 152 100
123 132 76 38 80 48 91 157 71 4 16 51 35 180 28
102 185 67 144 45 73 140 53 52 179 177 131 36 33
143 9 54 174 8 164 136 184 178 29 55 101 59 1 3
169 50 93 47 147 130 159 170 153 193 78 88 107 57
182
```

Candidate moves (k = 15)

min=74831 max=81611 avg=77384.93

best = 74831

TIME (ms)

avg = 73.65

min = 54.79 max = 110.13

Best solution nodes:

```
33 36 131 99 177 52 179 140 53 144 67 16 51 139
112 71 157 91 80 76 4 132 90 83 123 100 152 108 48
31 60 82 49 11 63 149 116 168 127 97 161 171 2 187
30 180 141 87 102 28 35 156 185 45 73 135 181 192
158 58 84 134 72 176 150 106 43 89 113 42 151 182
57 107 88 78 193 153 159 130 47 147 93 3 50 169 1
178 55 101 29 163 136 164 142 85 174 8 54 9 33
```

4. Best solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 20)

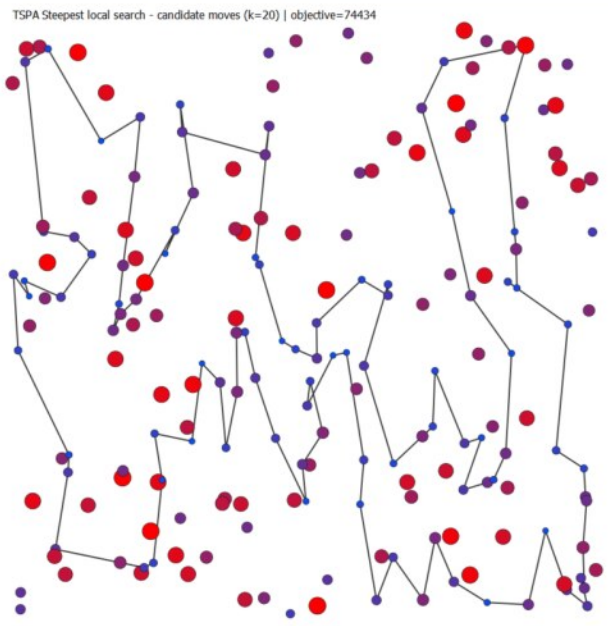
min=74434 max=80540 avg=77336.46

best = 74434

TIME (ms)

avg = 93.56

min = 76.23 max = 131.95



Best solution nodes:

```
176 72 134 131 9 54 33 52 179 144 67 16 51 112 139
71 157 4 132 76 90 83 160 123 100 38 91 80 48 31
82 60 108 152 49 149 63 116 168 97 127 69 171 2
187 30 141 180 28 87 102 156 35 185 45 135 73 41
53 140 177 36 84 58 192 158 111 138 182 57 64 107
88 78 153 193 75 170 146 159 130 47 3 50 169 1 178
29 136 186 164 85 93 113 89 42 151 106 150 174 176
```

Candidate moves (k = 30)

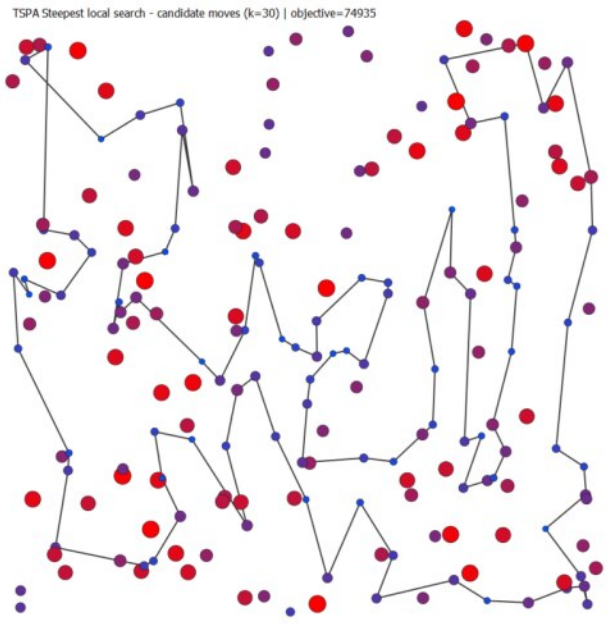
min=74935 max=80342 avg=77355.34

best = 74935

TIME (ms)

avg = 136.80

min = 110.79 max = 194.05



Best solution nodes:

```
134 84 73 140 53 177 36 131 54 9 33 52 179 144 67
16 51 35 28 180 90 83 123 160 100 132 76 71 4 157
91 80 31 48 60 108 152 49 149 63 116 168 97 127 69
171 2 189 187 30 141 98 87 102 185 45 135 181 58
158 192 182 57 64 88 193 78 153 107 146 170 159
130 47 147 21 55 101 29 136 184 178 1 169 3 50 93
43 113 89 42 151 106 150 85 142 164 8 174 176 134
```

Candidate moves (k = 50)

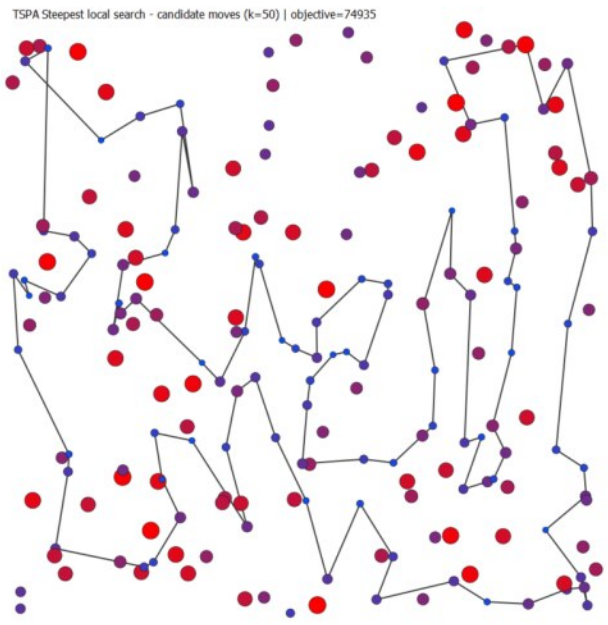
min=74935 max=80342 avg=77353.41

best = 74935

TIME (ms)

avg = 198.19

min = 161.17 max = 268.11



Best solution nodes:

```
134 84 73 140 53 177 36 131 54 9 33 52 179 144 67
16 51 35 28 180 90 83 123 160 100 132 76 71 4 157
91 80 31 48 60 108 152 49 149 63 116 168 97 127 69
171 2 189 187 30 141 98 87 102 185 45 135 181 58
158 192 182 57 64 88 193 78 153 107 146 170 159
130 47 147 21 55 101 29 136 184 178 1 169 3 50 93
43 113 89 42 151 106 150 85 142 164 8 174 176 134
```

Full neighborhood (k = infinity)

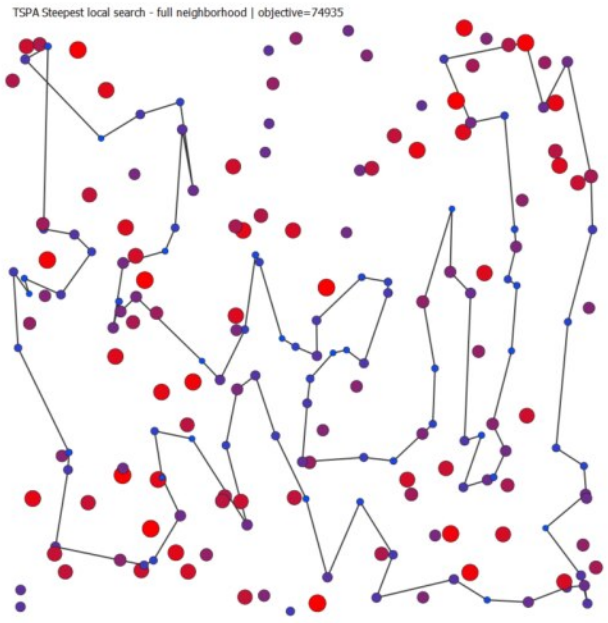
min=74935 max=80342 avg=77354.36

best = 74935

TIME (ms)

avg = 256.70

min = 207.59 max = 322.33



Best solution nodes:

```
134 84 73 140 53 177 36 131 54 9 33 52 179 144 67
16 51 35 28 180 90 83 123 160 100 132 76 71 4 157
91 80 31 48 60 108 152 49 149 63 116 168 97 127 69
171 2 189 187 30 141 98 87 102 185 45 135 181 58
158 192 182 57 64 88 193 78 153 107 146 170 159
130 47 147 21 55 101 29 136 184 178 1 169 3 50 93
43 113 89 42 151 106 150 85 142 164 8 174 176 134
```

4. Best solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 1)

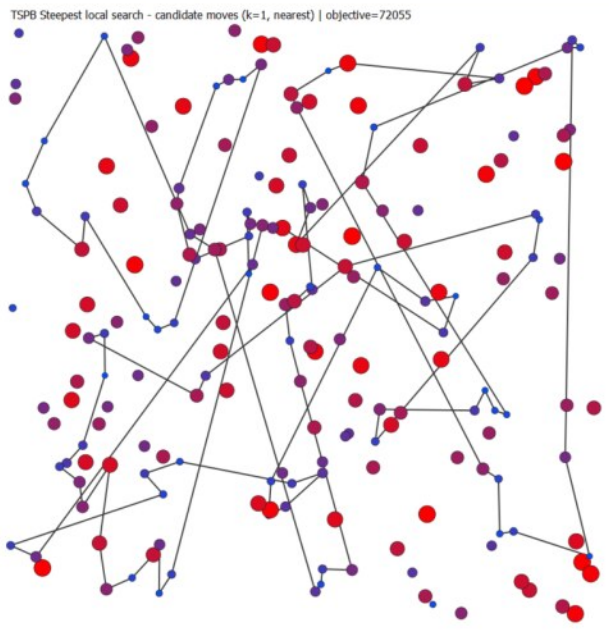
min=72055 max=90620 avg=81385.45

best = 72055

TIME (ms)

avg = 5.37

min = 4.09 max = 7.79



Best solution nodes:

```
174 160 69 133 136 172 21 176 191 130 121 198 96
48 47 140 154 158 138 145 28 149 116 30 189 82 87
179 88 98 151 122 173 184 3 72 46 109 190 194 70
44 89 43 156 182 153 20 14 110 118 94 105 81 57 79
90 129 127 80 66 67 10 114 196 75 64 33 177 125
170 6 29 132 197 36 113 91 62 175 142 157 134 45 8
117 95 192 54 162 76 59 85 51 181 167 40 71 60 1
174
```

Candidate moves (k = 5)

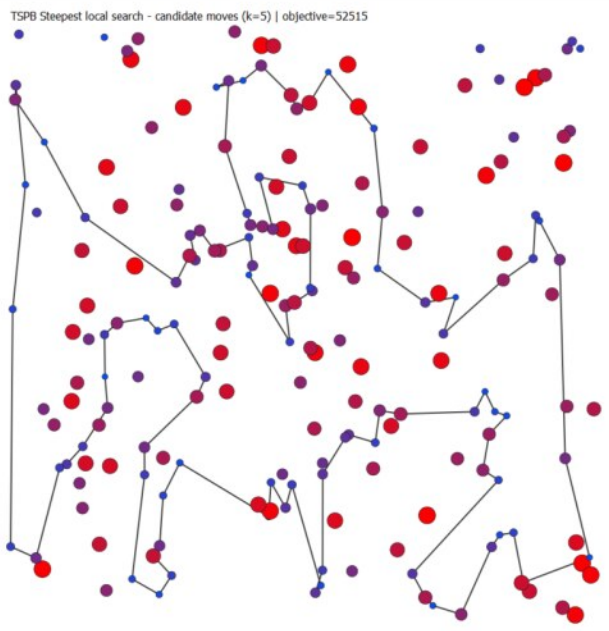
min=52515 max=58105 avg=54893.31

best = 52515

TIME (ms)

avg = 25.10

min = 19.05 max = 39.89



Best solution nodes:

```
155 71 175 134 45 142 157 60 40 85 59 51 167 87 82
179 181 34 24 67 66 84 80 127 129 90 52 182 153 99
39 171 17 74 20 14 2 110 118 148 114 196 10 78 192
54 162 76 185 57 89 32 156 43 48 96 121 198 63 133
136 8 117 120 72 46 184 3 173 122 151 69 160 172
189 183 30 21 176 38 158 149 13 146 28 65 1 174
132 29 6 115 170 125 26 154 140 47 64 33 155
```

Candidate moves (k = 10)

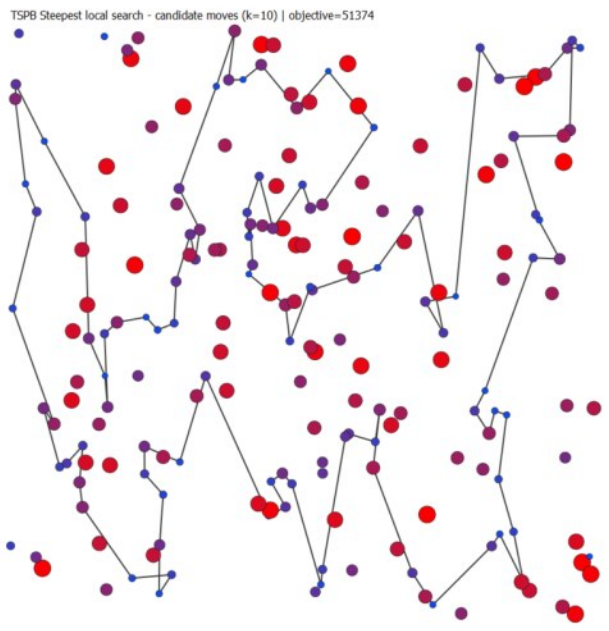
min=51374 max=56520 avg=53573.29

best = 51374

TIME (ms)

avg = 46.35

min = 37.35 max = 99.06



Best solution nodes:

```
70 190 54 162 192 104 76 3 184 151 122 69 160 136
172 133 120 117 72 46 86 57 89 156 48 96 198 58
121 130 183 21 30 38 47 140 154 26 125 115 170 177
158 149 146 13 28 145 65 103 178 132 29 6 197 36
175 45 134 157 60 71 155 40 64 85 51 59 73 167 87
179 34 24 67 66 31 99 171 74 20 2 14 153 90 129 52
80 127 10 148 196 114 187 139 61 94 105 81 169 70
```

Candidate moves (k = 15)

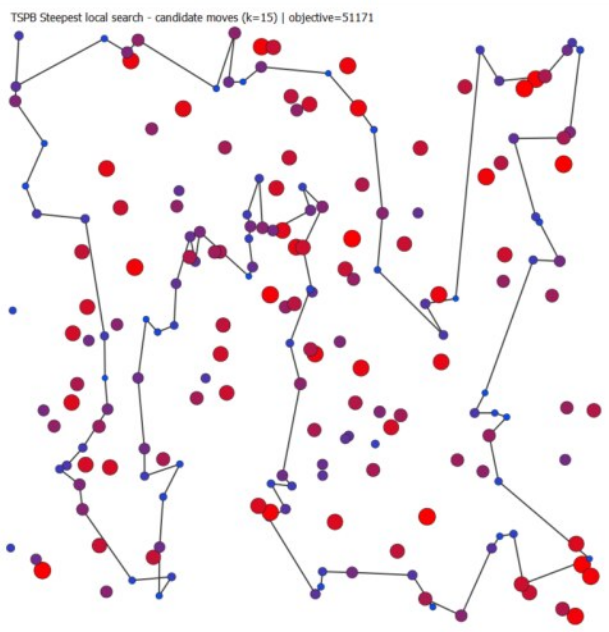
min=51171 max=56408 avg=53472.66

best = 51171

TIME (ms)

avg = 65.64

min = 50.07 max = 96.86



Best solution nodes:

```
190 70 169 81 94 105 61 139 187 196 114 148 10 127
80 129 90 52 153 110 2 14 20 74 17 39 171 99 88
179 87 82 85 51 59 167 73 98 151 184 3 137 86 72
46 117 8 120 133 136 172 160 69 183 21 30 38 47
140 154 97 155 71 40 60 157 134 45 175 36 197 132
29 6 115 170 125 158 145 28 149 13 37 146 116 147
42 121 58 198 96 48 89 83 57 185 76 192 162 54 190
```

4. Best solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 20)

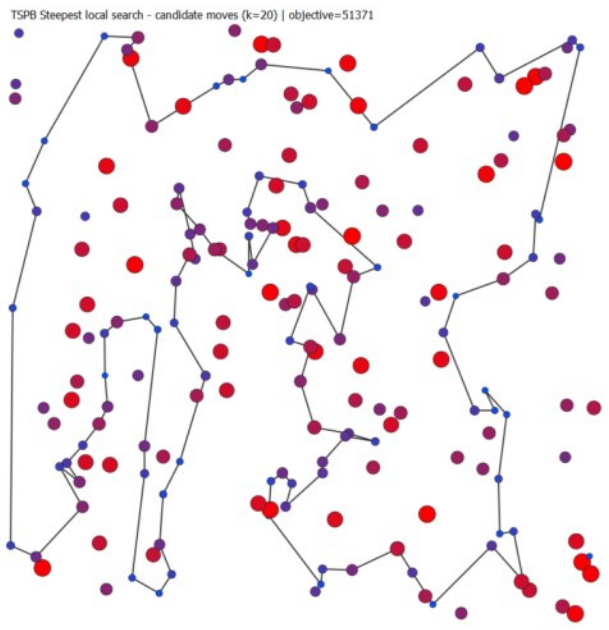
min=51371 max=56469 avg=53461.22

best = 51371

TIME (ms)

avg = 84.62

min = 67.94 max = 182.77



Best solution nodes:

```
180 115 170 125 26 154 140 155 71 175 134 45 157
60 40 64 47 38 176 21 30 130 191 183 189 69 172
160 117 8 136 133 120 72 46 76 135 27 3 184 151
111 98 25 24 67 34 152 181 51 167 73 59 85 87 179
88 163 99 171 74 2 14 20 153 90 127 129 80 192 54
78 10 196 114 105 94 70 190 57 89 48 96 198 121 50
147 42 116 149 28 145 65 1 174 36 132 197 29 6 180
```

Candidate moves (k = 30)

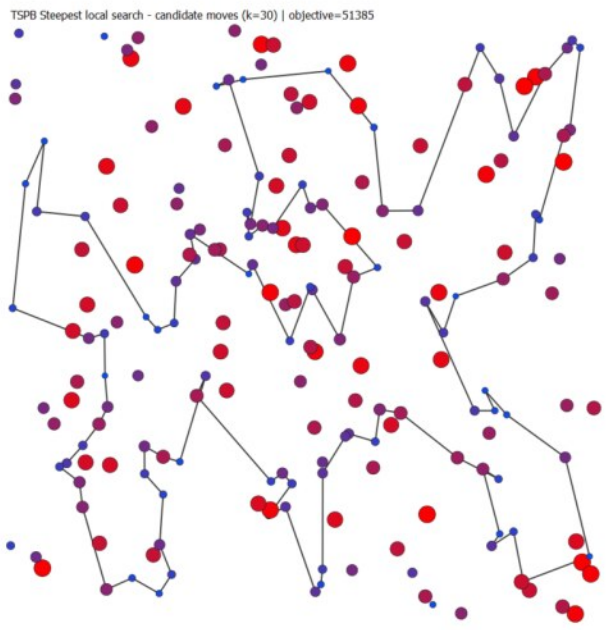
min=51385 max=56469 avg=53515.31

best = 51385

TIME (ms)

avg = 118.16

min = 90.92 max = 210.63



Best solution nodes:

```
96 121 198 120 136 133 172 8 117 72 46 86 76 135
27 3 184 151 160 69 30 21 38 47 140 154 158 145
149 28 65 177 125 170 115 180 6 29 132 197 36 62
175 134 45 157 60 71 155 141 40 64 33 59 73 167 85
51 82 87 179 181 34 24 67 66 84 150 153 182 20 74
14 2 110 118 90 127 129 80 162 192 54 78 10 196
114 139 61 105 94 81 169 187 70 190 104 185 57 89
96
```

Candidate moves (k = 50)

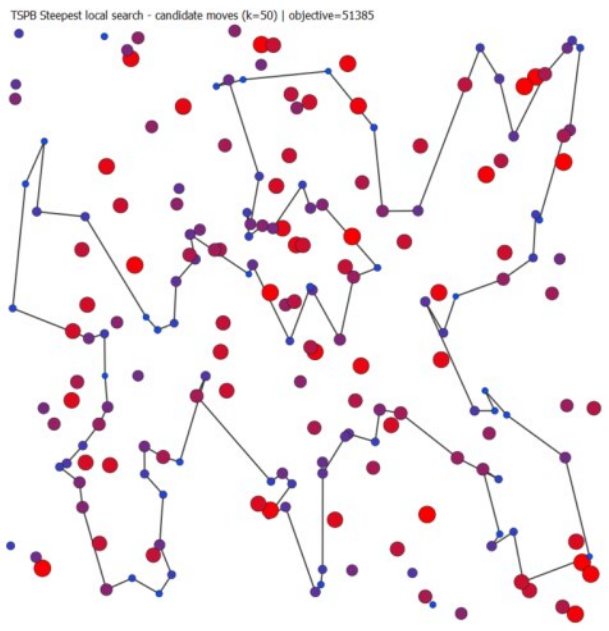
min=51385 max=56469 avg=53506.79

best = 51385

TIME (ms)

avg = 181.67

min = 144.40 max = 283.81



Best solution nodes:

```
96 121 198 120 136 133 172 8 117 72 46 86 76 135
27 3 184 151 160 69 30 21 38 47 140 154 158 145
149 28 65 177 125 170 115 180 6 29 132 197 36 62
175 134 45 157 60 71 155 141 40 64 33 59 73 167 85
51 82 87 179 181 34 24 67 66 84 150 153 182 20 74
14 2 110 118 90 127 129 80 162 192 54 78 10 196
114 139 61 105 94 81 169 187 70 190 104 185 57 89
96
```

Full neighborhood (k = infinity)

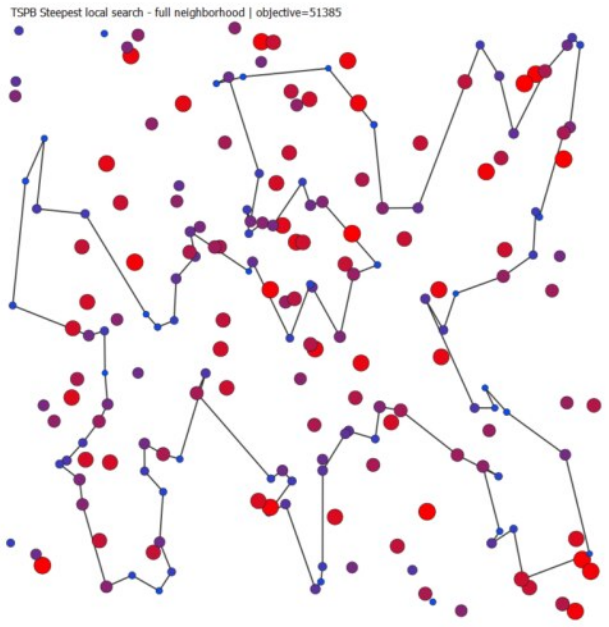
min=51385 max=56469 avg=53505.13

best = 51385

TIME (ms)

avg = 262.19

min = 211.10 max = 339.42



Best solution nodes:

```
96 121 198 120 136 133 172 8 117 72 46 86 76 135
27 3 184 151 160 69 30 21 38 47 140 154 158 145
149 28 65 177 125 170 115 180 6 29 132 197 36 62
175 134 45 157 60 71 155 141 40 64 33 59 73 167 85
51 82 87 179 181 34 24 67 66 84 150 153 182 20 74
14 2 110 118 90 127 129 80 162 192 54 78 10 196
114 139 61 105 94 81 169 187 70 190 104 185 57 89
96
```

5. Worst solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 1)

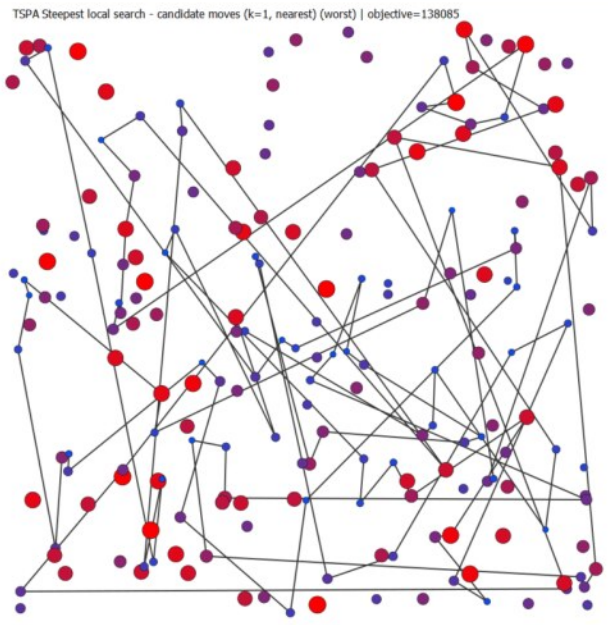
min=108540 max=138085 avg=119836.08

worst = 138085

TIME (ms)

avg = 5.93

min = 3.96 max = 30.61



Worst solution nodes:

```
101 24 197 147 21 42 89 164 8 30 23 149 63 168 69
95 97 127 180 28 189 114 135 50 1 169 144 67 185
31 48 152 2 187 171 71 157 0 91 80 38 14 83 123
160 29 178 184 186 195 136 161 129 88 34 155 26
193 153 10 141 87 27 170 53 177 33 36 131 106 150
41 61 51 16 181 158 93 47 182 57 138 165 117 65
134 58 84 132 76 45 172 156 35 72 176 174 107 130
15 101
```

Candidate moves (k = 5)

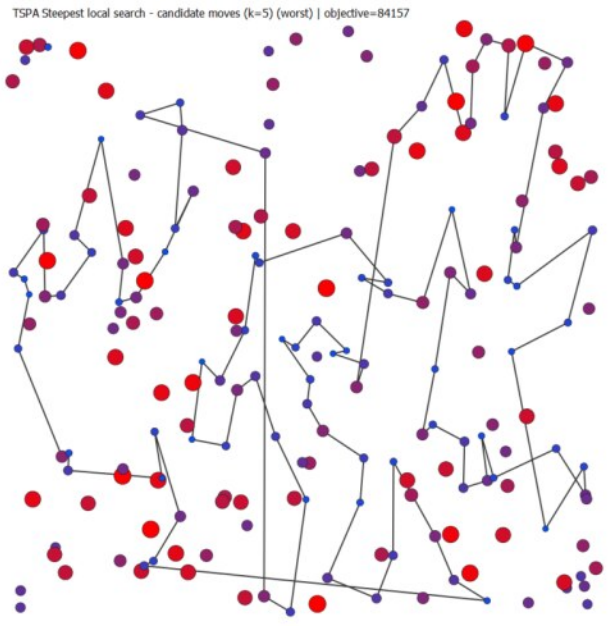
min=74889 max=84157 avg=79905.04

worst = 84157

TIME (ms)

avg = 27.42

min = 22.80 max = 41.21



Worst solution nodes:

```
163 19 178 29 55 101 169 1 3 50 147 47 93 107 159
146 130 89 106 42 151 150 176 72 174 142 85 164
194 8 54 33 9 143 16 51 35 28 180 141 87 102 185
45 135 114 25 139 91 157 71 76 4 132 90 123 100 80
108 152 49 11 60 116 149 63 168 95 97 127 187 30
189 2 171 57 182 138 134 158 192 181 58 84 41 140
53 67 144 52 36 177 131 99 34 186 136 56 184 24
163
```

Candidate moves (k = 10)

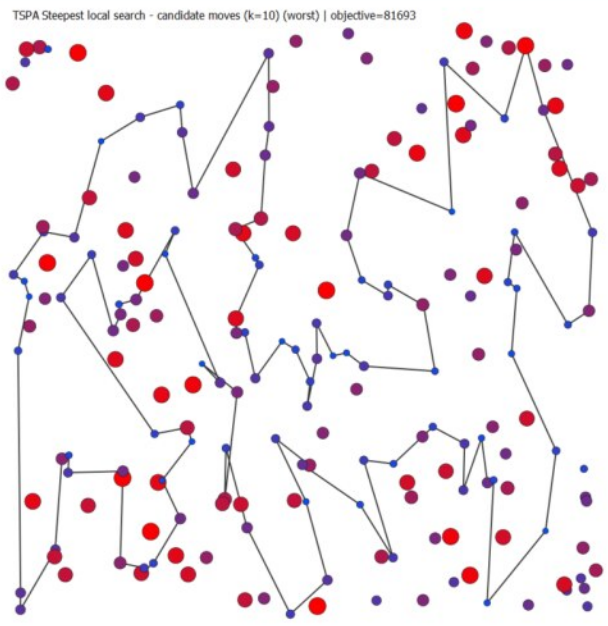
min=75168 max=81693 avg=77589.12

worst = 81693

TIME (ms)

avg = 52.49

min = 41.16 max = 101.95



Worst solution nodes:

```
93 130 107 57 89 42 106 151 150 176 134 84 158 58
73 61 45 135 181 114 98 87 27 102 180 28 132 76 90
123 83 160 152 49 30 145 141 187 189 2 32 171 92
161 127 97 95 94 69 104 129 168 63 149 116 60 108
80 91 157 71 4 115 112 139 133 196 51 16 172 156
35 185 67 144 53 140 179 52 177 36 131 174 8 9 54
33 143 62 164 136 178 29 101 147 37 47 1 3 50 93
```

Candidate moves (k = 15)

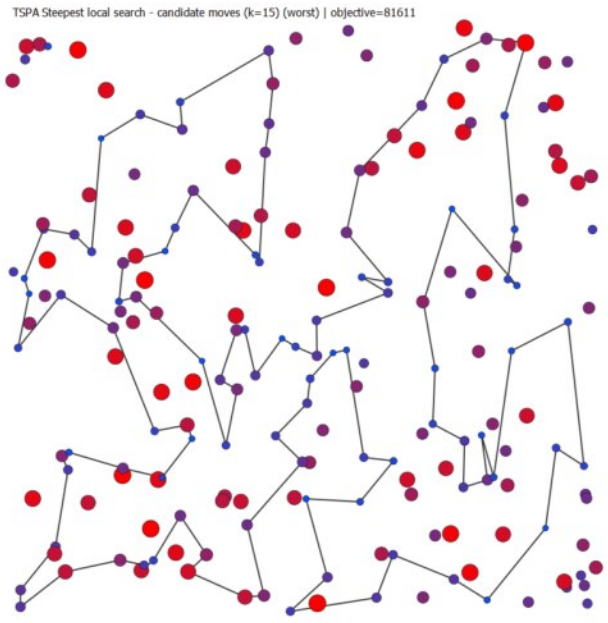
min=74831 max=81611 avg=77384.93

worst = 81611

TIME (ms)

avg = 73.65

min = 54.79 max = 110.13



Worst solution nodes:

```
157 71 91 80 152 108 60 149 63 168 49 160 30 145
141 187 161 97 95 127 69 129 104 110 92 171 32 2
189 10 183 22 25 98 73 45 140 53 177 36 84 134 58
135 181 114 192 158 182 57 107 130 159 47 93 89
106 42 151 150 176 174 8 164 50 3 1 178 29 163 136
186 62 143 9 33 54 52 179 144 67 185 35 156 28 102
87 180 90 123 100 132 76 4 51 16 139 112 5 115 157
```

5. Worst solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 20)

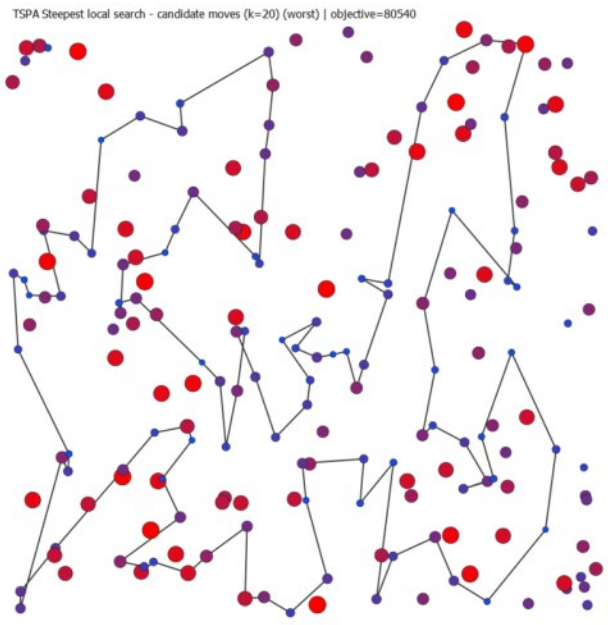
min=74434 max=80540 avg=77336.46

worst = 80540

TIME (ms)

avg = 93.56

min = 76.23 max = 131.95



Worst solution nodes:

```
157 71 91 80 152 108 60 49 63 149 116 168 97 127
95 104 129 69 161 30 145 141 187 189 92 171 32 2
183 10 98 22 25 114 181 135 73 84 58 134 192 158
138 182 57 107 130 93 106 89 151 42 150 176 72 174
8 164 50 3 1 178 29 163 136 186 9 33 54 131 99 36
177 179 144 52 67 53 140 45 185 156 35 102 87 28
180 90 123 83 100 132 76 4 51 16 139 112 5 115 157
```

Candidate moves (k = 30)

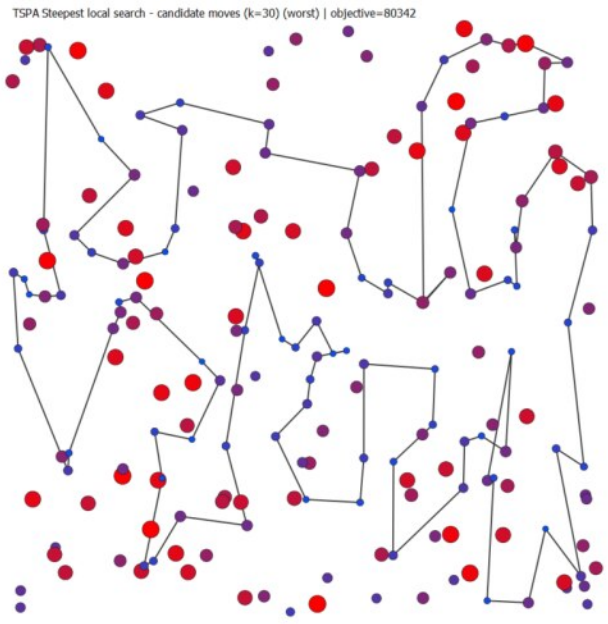
min=74935 max=80342 avg=77355.34

worst = 80342

TIME (ms)

avg = 136.80

min = 110.79 max = 194.05



Worst solution nodes:

```
123 83 160 97 127 95 168 116 149 63 11 49 60 48 80
38 108 152 100 132 76 71 91 157 112 139 62 143 33
54 9 194 142 8 186 136 163 19 29 55 17 101 178 184
164 85 3 50 1 169 59 46 21 147 47 159 130 153 107
64 57 89 42 93 113 106 150 151 158 134 72 176 174
131 84 58 135 45 140 53 179 36 177 52 144 67 51 16
35 87 98 189 2 171 187 30 141 28 180 90 123
```

Candidate moves (k = 50)

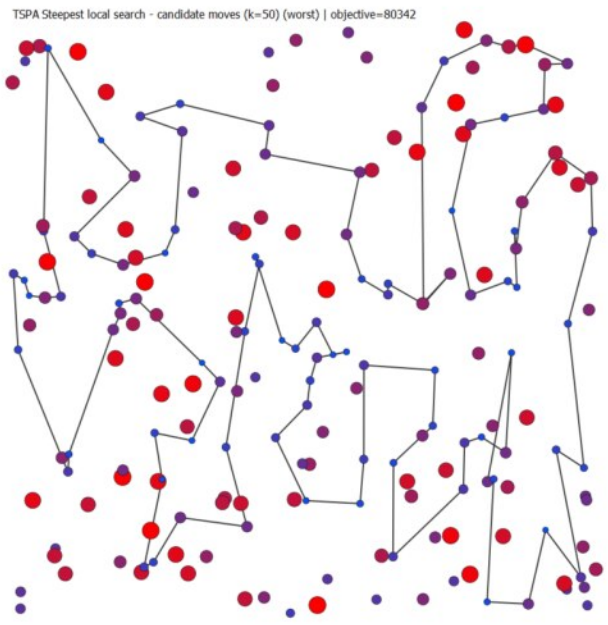
min=74935 max=80342 avg=77353.41

worst = 80342

TIME (ms)

avg = 198.19

min = 161.17 max = 268.11



Worst solution nodes:

```
123 83 160 97 127 95 168 116 149 63 11 49 60 48 80
38 108 152 100 132 76 71 91 157 112 139 62 143 33
54 9 194 142 8 186 136 163 19 29 55 17 101 178 184
164 85 3 50 1 169 59 46 21 147 47 159 130 153 107
64 57 89 42 93 113 106 150 151 158 134 72 176 174
131 84 58 135 45 140 53 179 36 177 52 144 67 51 16
35 87 98 189 2 171 187 30 141 28 180 90 123
```

Full neighborhood (k = infinity)

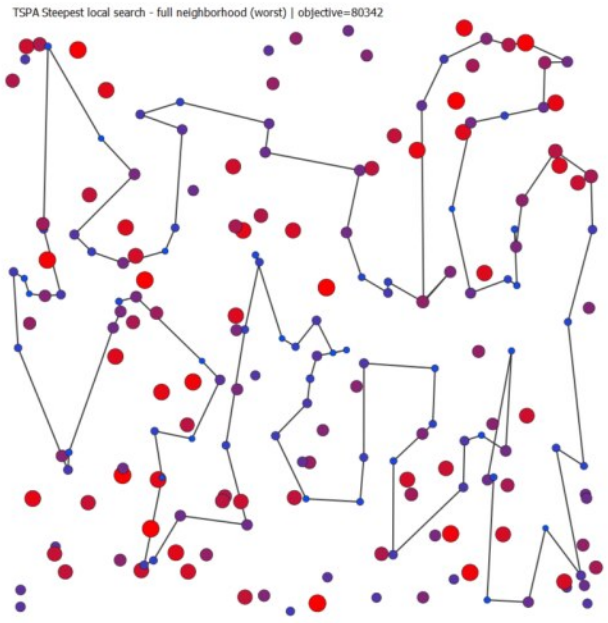
min=74935 max=80342 avg=77354.36

worst = 80342

TIME (ms)

avg = 256.70

min = 207.59 max = 322.33



Worst solution nodes:

```
123 83 160 97 127 95 168 116 149 63 11 49 60 48 80
38 108 152 100 132 76 71 91 157 112 139 62 143 33
54 9 194 142 8 186 136 163 19 29 55 17 101 178 184
164 85 3 50 1 169 59 46 21 147 47 159 130 153 107
64 57 89 42 93 113 106 150 151 158 134 72 176 174
131 84 58 135 45 140 53 179 36 177 52 144 67 51 16
35 87 98 189 2 171 187 30 141 28 180 90 123
```

5. Worst solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 1)

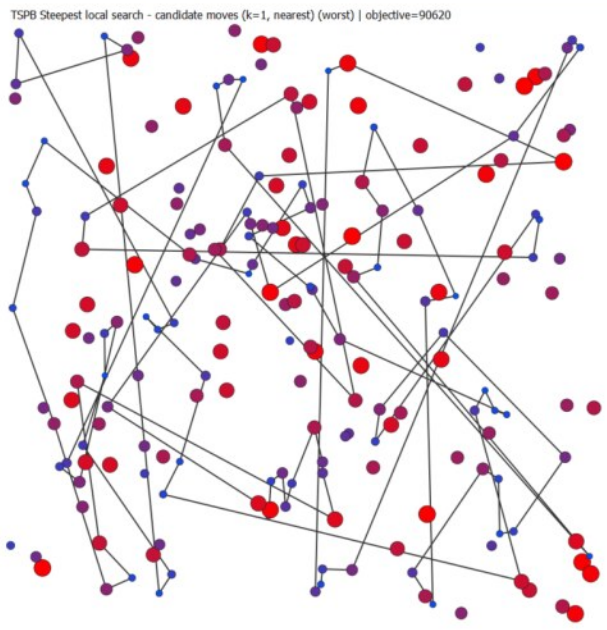
min=72055 max=90620 avg=81385.45

worst = 90620

TIME (ms)

avg = 5.37

min = 4.09 max = 7.79



Worst solution nodes:

```
82 89 44 193 120 168 189 11 43 158 138 10 114 196
67 66 192 118 14 20 153 182 99 171 162 54 104 57
79 185 76 135 110 68 63 121 198 96 29 132 197 26
125 170 6 45 134 116 147 146 37 47 140 154 64 40
60 2 188 80 127 129 90 27 184 3 172 46 72 160 69
21 149 28 145 65 103 62 175 91 19 126 181 152 25
167 51 73 59 85 107 115 133 128 187 105 94 88 179
87 82
```

Candidate moves (k = 5)

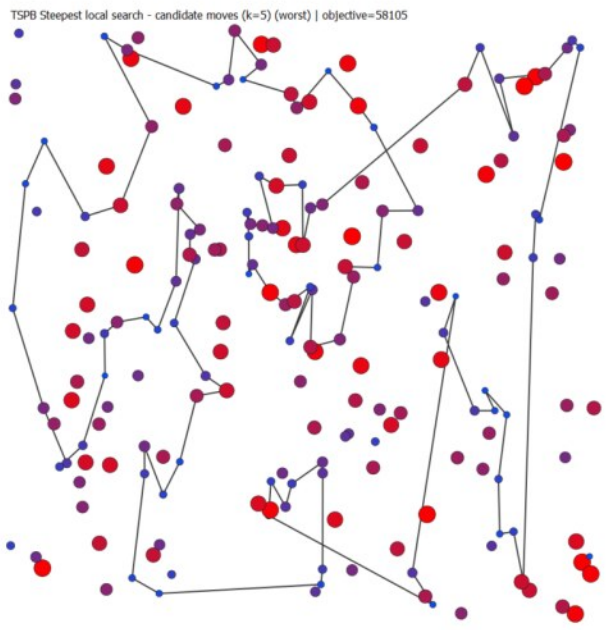
min=52515 max=58105 avg=54893.31

worst = 58105

TIME (ms)

avg = 25.10

min = 19.05 max = 39.89



Worst solution nodes:

```
147 121 198 58 48 96 43 156 89 83 57 104 185 76 75
135 27 111 3 151 184 122 128 160 69 172 133 136 8
117 120 92 72 137 46 86 194 190 187 70 169 81 94
105 114 196 10 2 14 20 153 90 127 129 80 192 54 99
171 85 59 51 167 152 181 179 87 134 175 71 155 60
40 33 49 64 47 21 176 30 183 191 130 38 140 154 26
125 170 6 132 29 178 65 28 149 158 15 50 116 147
```

Candidate moves (k = 10)

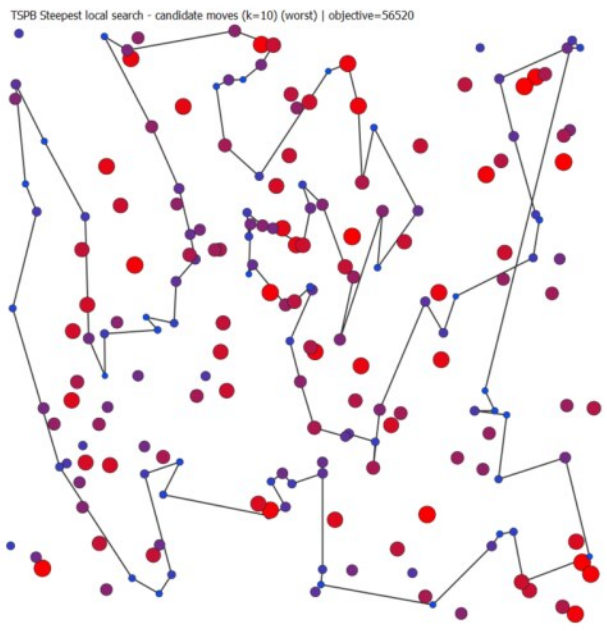
min=51374 max=56520 avg=53573.29

worst = 56520

TIME (ms)

avg = 46.35

min = 37.35 max = 99.06



Worst solution nodes:

```
83 79 57 104 76 185 27 135 75 86 72 46 137 117 8
133 172 136 69 160 122 173 184 3 151 98 25 34 24
67 31 66 162 192 54 10 114 196 187 70 81 105 94
127 129 80 90 153 118 110 2 14 20 74 171 87 179
152 181 167 73 59 51 85 60 40 71 45 134 175 132 65
145 28 146 149 158 177 170 125 140 154 47 38 21 30
130 50 116 147 58 164 48 96 198 121 63 120 89 44
83
```

Candidate moves (k = 15)

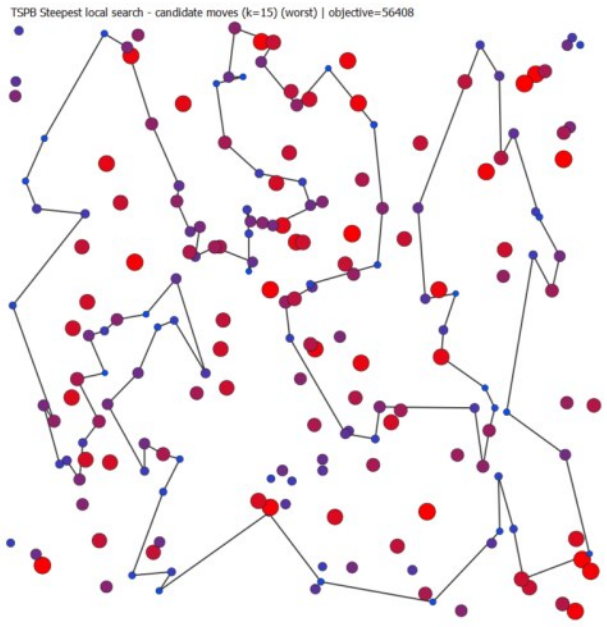
min=51171 max=56408 avg=53472.66

worst = 56408

TIME (ms)

avg = 65.64

min = 50.07 max = 96.86



Worst solution nodes:

```
52 182 80 66 67 34 151 122 3 184 76 185 57 89 156
48 164 58 198 96 121 63 120 72 46 86 117 8 136 133
172 69 160 189 21 183 30 191 130 50 147 116 149 28
145 158 65 103 178 132 29 197 6 180 19 170 177 125
26 154 38 64 47 140 97 115 71 155 141 40 60 175 45
134 85 87 171 20 153 14 2 110 118 90 10 143 148
196 114 187 18 70 190 104 162 54 192 106 127 129
52
```

5. Worst solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Candidate moves (k = 20)

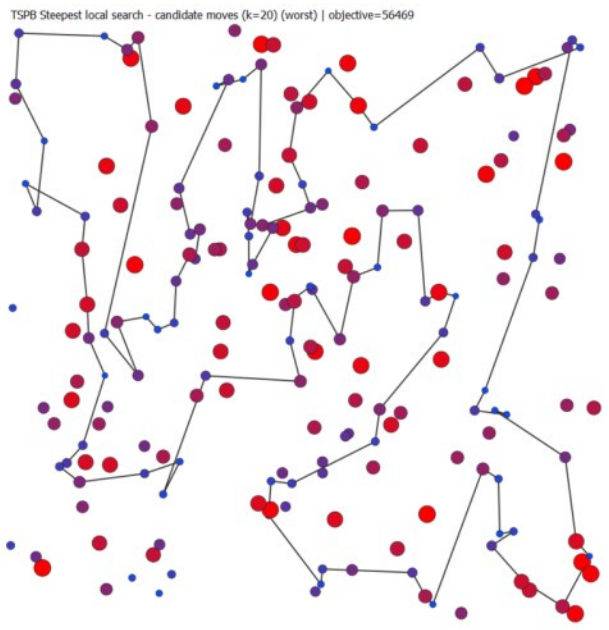
min=51371 max=56469 avg=53461.22

worst = 56469

TIME (ms)

avg = 84.62

min = 67.94 max = 182.77



Worst solution nodes:

```
183 30 130 198 121 96 48 120 136 133 172 69 160
117 8 86 46 72 166 156 89 83 57 190 70 105 94 196
114 10 127 80 90 129 118 68 110 93 2 74 14 20 153
182 171 39 99 179 87 85 59 167 67 66 192 54 159
162 104 185 76 135 27 3 184 122 173 151 98 64 60
40 71 197 132 29 6 170 177 55 138 158 28 145 149
146 37 116 147 42 50 125 97 26 154 140 47 38 176
21 183
```

Candidate moves (k = 30)

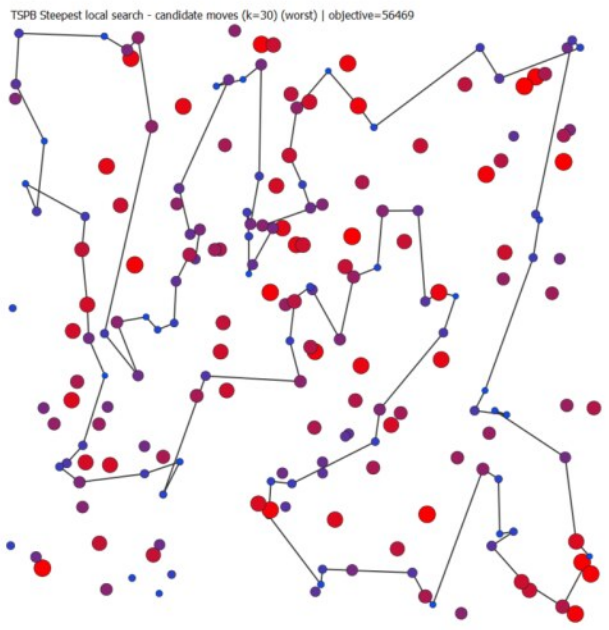
min=51385 max=56469 avg=53515.31

worst = 56469

TIME (ms)

avg = 118.16

min = 90.92 max = 210.63



Worst solution nodes:

```
183 30 130 198 121 96 48 120 136 133 172 69 160
117 8 86 46 72 166 156 89 83 57 190 70 105 94 196
114 10 127 80 90 129 118 68 110 93 2 74 14 20 153
182 171 39 99 179 87 85 59 167 67 66 192 54 159
162 104 185 76 135 27 3 184 122 173 151 98 64 60
40 71 197 132 29 6 170 177 55 138 158 28 145 149
146 37 116 147 42 50 125 97 26 154 140 47 38 176
21 183
```

Candidate moves (k = 50)

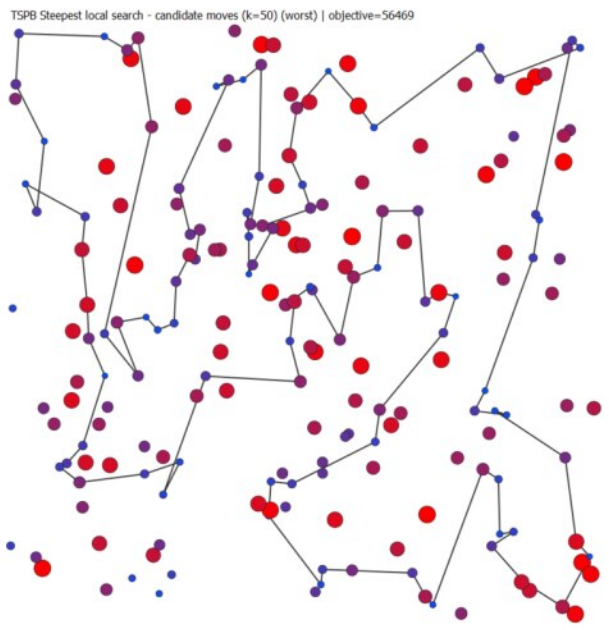
min=51385 max=56469 avg=53506.79

worst = 56469

TIME (ms)

avg = 181.67

min = 144.40 max = 283.81



Worst solution nodes:

```
183 30 130 198 121 96 48 120 136 133 172 69 160
117 8 86 46 72 166 156 89 83 57 190 70 105 94 196
114 10 127 80 90 129 118 68 110 93 2 74 14 20 153
182 171 39 99 179 87 85 59 167 67 66 192 54 159
162 104 185 76 135 27 3 184 122 173 151 98 64 60
40 71 197 132 29 6 170 177 55 138 158 28 145 149
146 37 116 147 42 50 125 97 26 154 140 47 38 176
21 183
```

Full neighborhood (k = infinity)

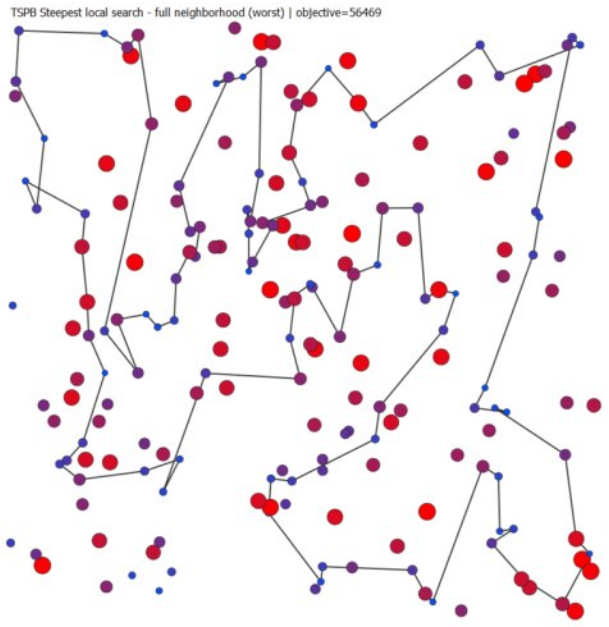
min=51385 max=56469 avg=53505.13

worst = 56469

TIME (ms)

avg = 262.19

min = 211.10 max = 339.42



Worst solution nodes:

```
183 30 130 198 121 96 48 120 136 133 172 69 160
117 8 86 46 72 166 156 89 83 57 190 70 105 94 196
114 10 127 80 90 129 118 68 110 93 2 74 14 20 153
182 171 39 99 179 87 85 59 167 67 66 192 54 159
162 104 185 76 135 27 3 184 122 173 151 98 64 60
40 71 197 132 29 6 170 177 55 138 158 28 145 149
146 37 116 147 42 50 125 97 26 154 140 47 38 176
21 183
```

6. Conclusions

Candidate moves substantially reduce the time needed by steepest local search because the algorithm evaluates only a small directly generated subset of the full neighborhood. Increasing the candidate-list size generally improves solution quality, but also increases the number of moves evaluated and therefore the runtime.

Main conclusions

For TSPA, the lowest average objective was obtained by Candidate moves (k = 20) (77336.46). For TSPB, the best average was obtained by Candidate moves (k = 20) (53461.22).

Runtime followed the expected pruning trade-off: candidate moves with k = 1 were fastest in both TSPA and TSPB, while the full-neighborhood baseline was the slowest. Increasing K made the candidate neighborhood larger and generally increased the average runtime.

The solution-quality trend went in the opposite direction. The single-nearest case had averages 119836.08 (TSPA) and 81385.45 (TSPB). Increasing K improved the averages strongly up to k = 20, which was the best average setting for both instances; k = 30 and k = 50 then stayed close to the full-neighborhood quality, showing a plateau rather than unlimited improvement.

The best individual routes did not have to come from k = infinity: TSPA's lowest observed value came from Candidate moves (k = 20) (74434), and TSPB's came from Candidate moves (k = 15) (51171). A plausible explanation is that restricting the neighborhood changes the sequence of accepted moves and therefore the local optimum basin; one candidate-restricted run can avoid a full-neighborhood move sequence that ends in a worse local optimum.

All generated routes passed structural validation, and every final objective agreed with the accumulated direct move deltas.

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-04-candidate-moves

7. Additional experiment

Objective score: average (minimum - maximum); lower values are better.

Ranking policy	TSPA	TSPB
k = 1 - distance intra + costs inter	119836.08(108540 - 138085)	81385.45(72055 - 90620)
k = 1 - distance + costs intra/inter	112821.44(100511 - 129657)	77688.01(68614 - 87565)
k = 5 - distance intra + costs inter	79905.04(74889 - 84157)	54893.31(52515 - 58105)
k = 5 - distance + costs intra/inter	78271.01(75185 - 82014)	53743.36(51687 - 56293)
k = 10 - distance intra + costs inter	77589.12(75168 - 81693)	53573.29(51374 - 56520)
k = 10 - distance + costs intra/inter	77517.70(75153 - 80930)	**53432.75(51236 - 55791)**
k = 15 - distance intra + costs inter	77384.93(74831 - 81611)	53472.66(51171 - 56408)
k = 15 - distance + costs intra/inter	77330.48(74577 - 81333)	53548.49(51371 - 55822)
k = 20 - distance intra + costs inter	77336.46(74434 - 80540)	53461.22(51371 - 56469)
k = 20 - distance + costs intra/inter	**77327.07(75005 - 81952)**	53492.50(51385 - 56316)
k = 30 - distance intra + costs inter	77355.34(74935 - 80342)	53515.31(51385 - 56469)
k = 30 - distance + costs intra/inter	77331.75(74935 - 80341)	53517.09(51385 - 56469)
k = 50 - distance intra + costs inter	77353.41(74935 - 80342)	53506.79(51385 - 56469)
k = 50 - distance + costs intra/inter	77349.74(74935 - 80341)	53502.84(51385 - 56469)

Runtime: average (minimum - maximum) milliseconds; lower values are better.

Ranking policy	TSPA	TSPB
k = 1 - distance intra + costs inter	**3.69(2.82 - 6.09) ms**	**3.75(3.02 - 5.01) ms**
k = 1 - distance + costs intra/inter	6.40(4.94 - 10.68) ms	5.48(4.05 - 8.19) ms
k = 5 - distance intra + costs inter	17.57(14.55 - 24.05) ms	16.35(13.82 - 22.22) ms
k = 5 - distance + costs intra/inter	24.29(19.72 - 33.11) ms	20.46(17.28 - 29.79) ms
k = 10 - distance intra + costs inter	31.27(26.75 - 40.90) ms	28.75(24.27 - 39.70) ms
k = 10 - distance + costs intra/inter	42.08(35.00 - 57.16) ms	35.54(30.63 - 47.78) ms
k = 15 - distance intra + costs inter	44.47(34.84 - 122.42) ms	40.75(35.55 - 108.95) ms
k = 15 - distance + costs intra/inter	58.88(50.23 - 75.21) ms	48.51(41.53 - 62.95) ms
k = 20 - distance intra + costs inter	56.89(48.66 - 74.62) ms	51.11(43.31 - 67.86) ms
k = 20 - distance + costs intra/inter	74.90(63.31 - 96.55) ms	60.92(52.19 - 77.42) ms
k = 30 - distance intra + costs inter	81.33(68.71 - 130.76) ms	71.58(62.41 - 124.03) ms
k = 30 - distance + costs intra/inter	101.88(87.58 - 126.01) ms	82.13(70.09 - 119.00) ms
k = 50 - distance intra + costs inter	123.07(105.56 - 179.49) ms	109.83(95.64 - 171.84) ms
k = 50 - distance + costs intra/inter	148.22(127.82 - 218.16) ms	122.16(107.87 - 145.21) ms

lowest average objective lowest average runtime

Ranking-policy comparison

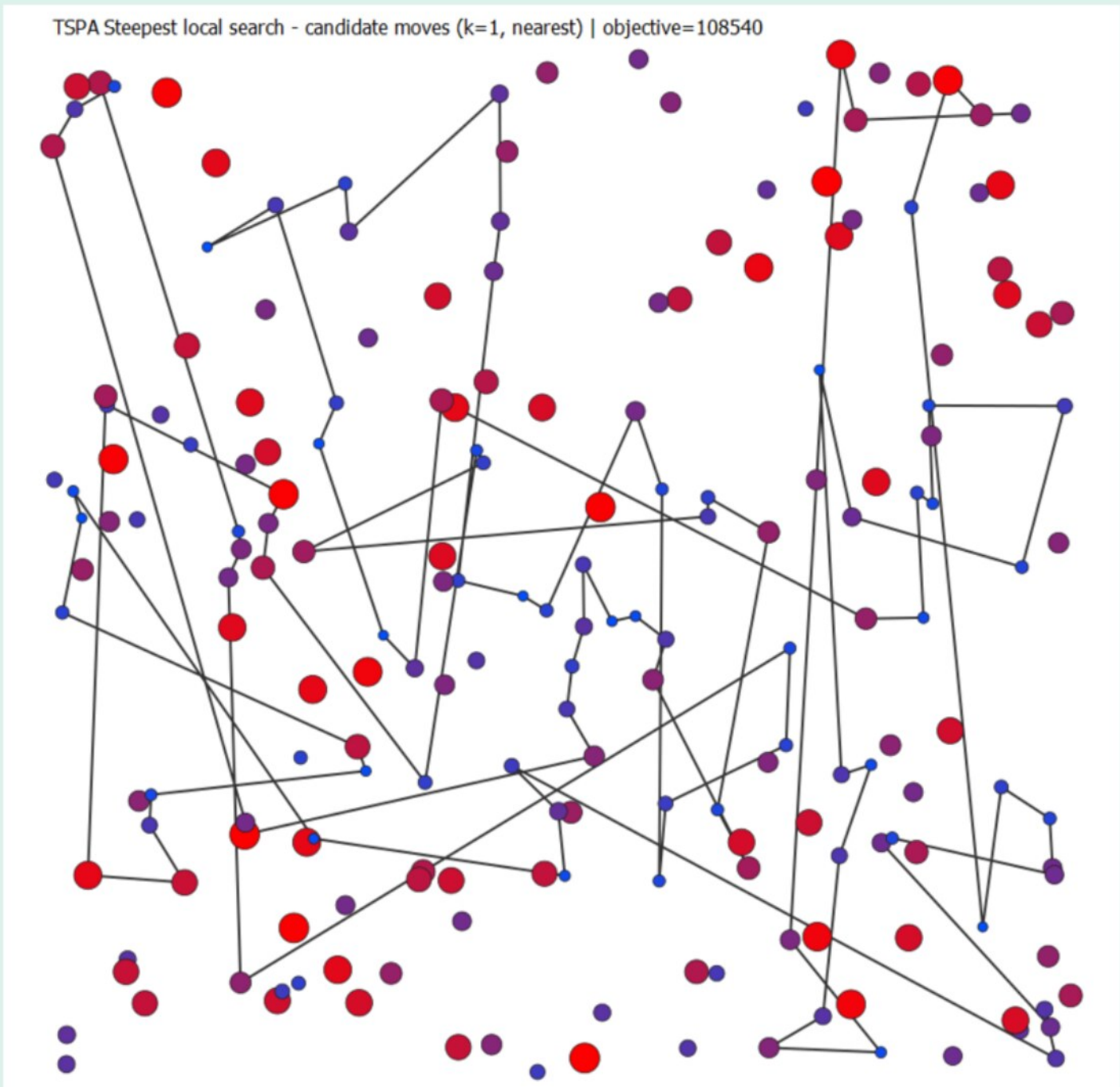
TSPA best objective: k = 20 - distance + costs intra/inter (77327.07).

TSPB best objective: k = 10 - distance + costs intra/inter (53432.75).

TSPA fastest average: k = 1 - distance intra + costs inter (3.69 ms).

TSPB fastest average: k = 1 - distance intra + costs inter (3.75 ms).

8. Large best visualizations - TSPA



BEST SOLUTION

Candidate moves (k = 1, nearest)

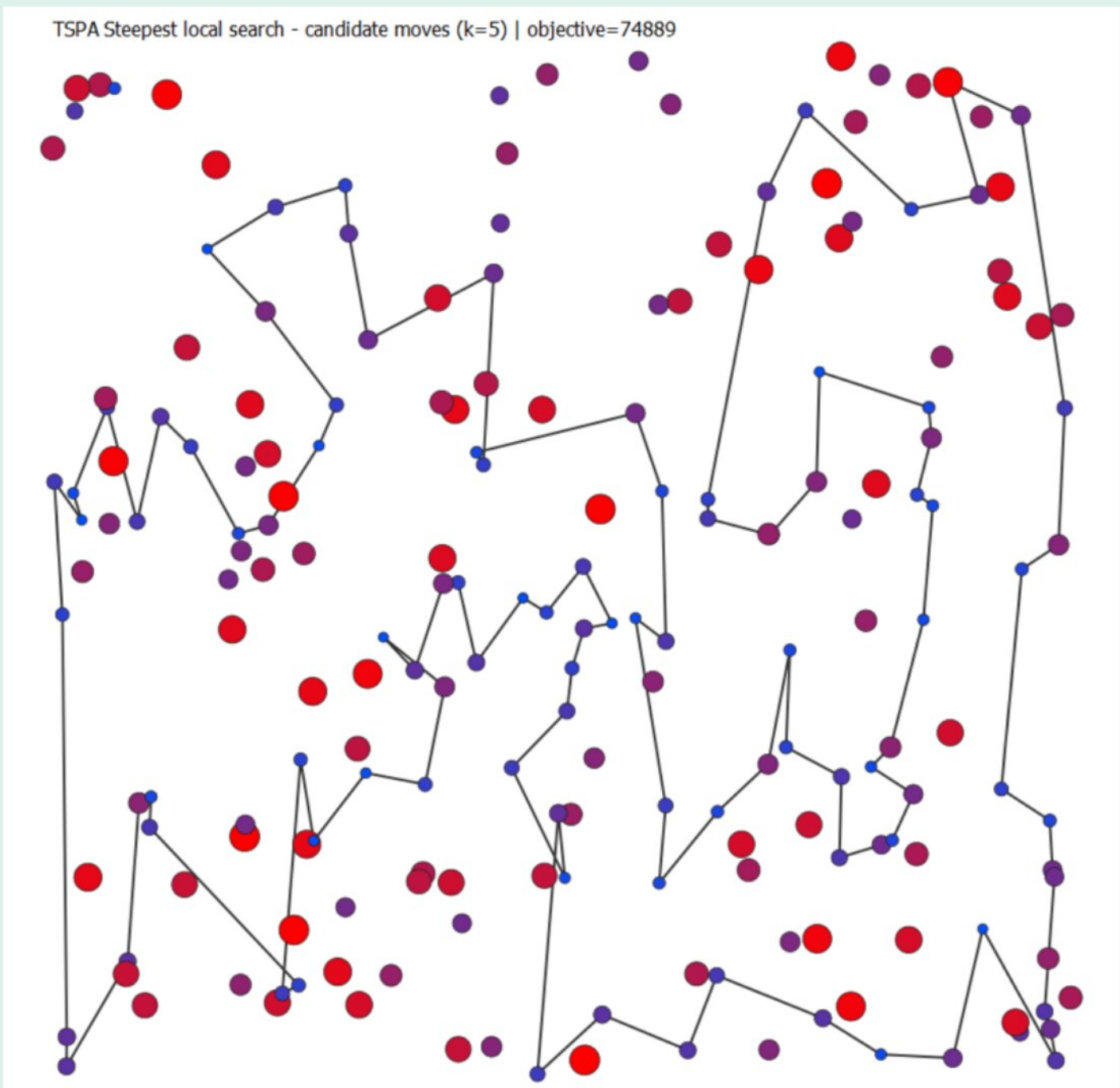
min=108540
max=138085
avg=119836.08

objective 108540

Best solution nodes:

157 71 115 112 139 35 67
144 143 33 58 84 176 174
92 160 83 123 7 48 31 77
161 125 41 140 53 179 52
177 36 131 99 65 134 194
8 9 54 162 16 51 87 148
90 68 60 82 188 120 127
97 141 145 168 63 149 187
135 73 45 78 153 193 42
89 170 146 159 130 107
178 29 17 55 24 197 138
57 111 182 151 106 150
164 85 47 147 1 50 3 93
81 196 28 180 132 76 91
80 157

100 selected nodes; closed route



BEST SOLUTION

Candidate moves (k = 5)

min=74889
max=84157
avg=79905.04

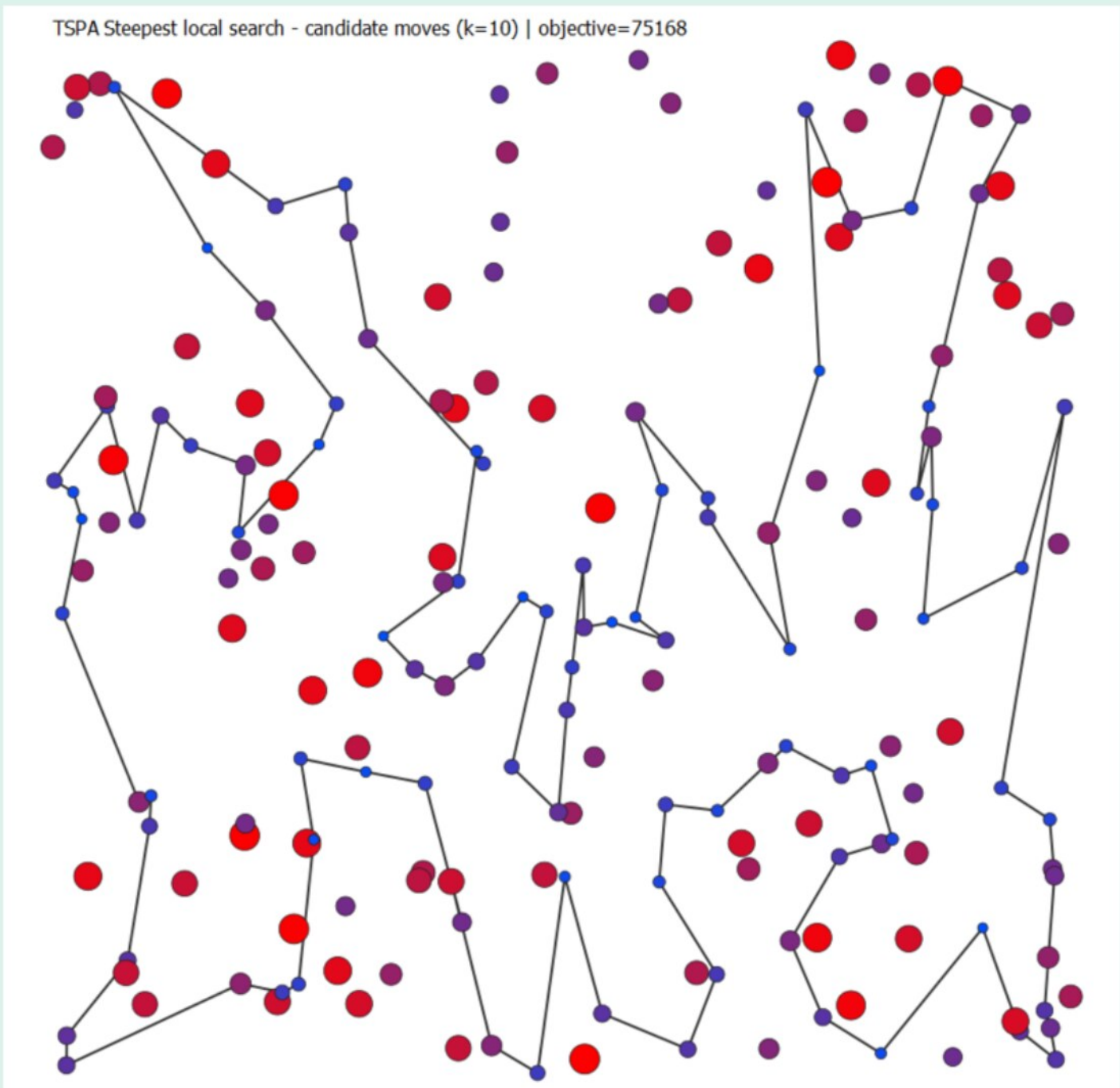
objective 74889

Best solution nodes:

164 1 169 3 50 93 43 106
113 89 42 151 150 176 174
72 134 58 84 36 131 33
143 51 16 139 4 71 157 91
80 38 76 132 90 123 152
108 49 60 149 63 116 168
129 104 69 95 97 127 2
171 30 187 141 87 102 180
28 156 35 185 67 144 52
177 179 53 140 45 135 73
114 181 192 158 182 57 64
107 78 193 153 170 146
159 130 47 37 147 55 29
101 178 136 186 9 54 8
142 164

100 selected nodes; closed route

8. Large best visualizations - TSPA



BEST SOLUTION

Candidate moves (k = 10)

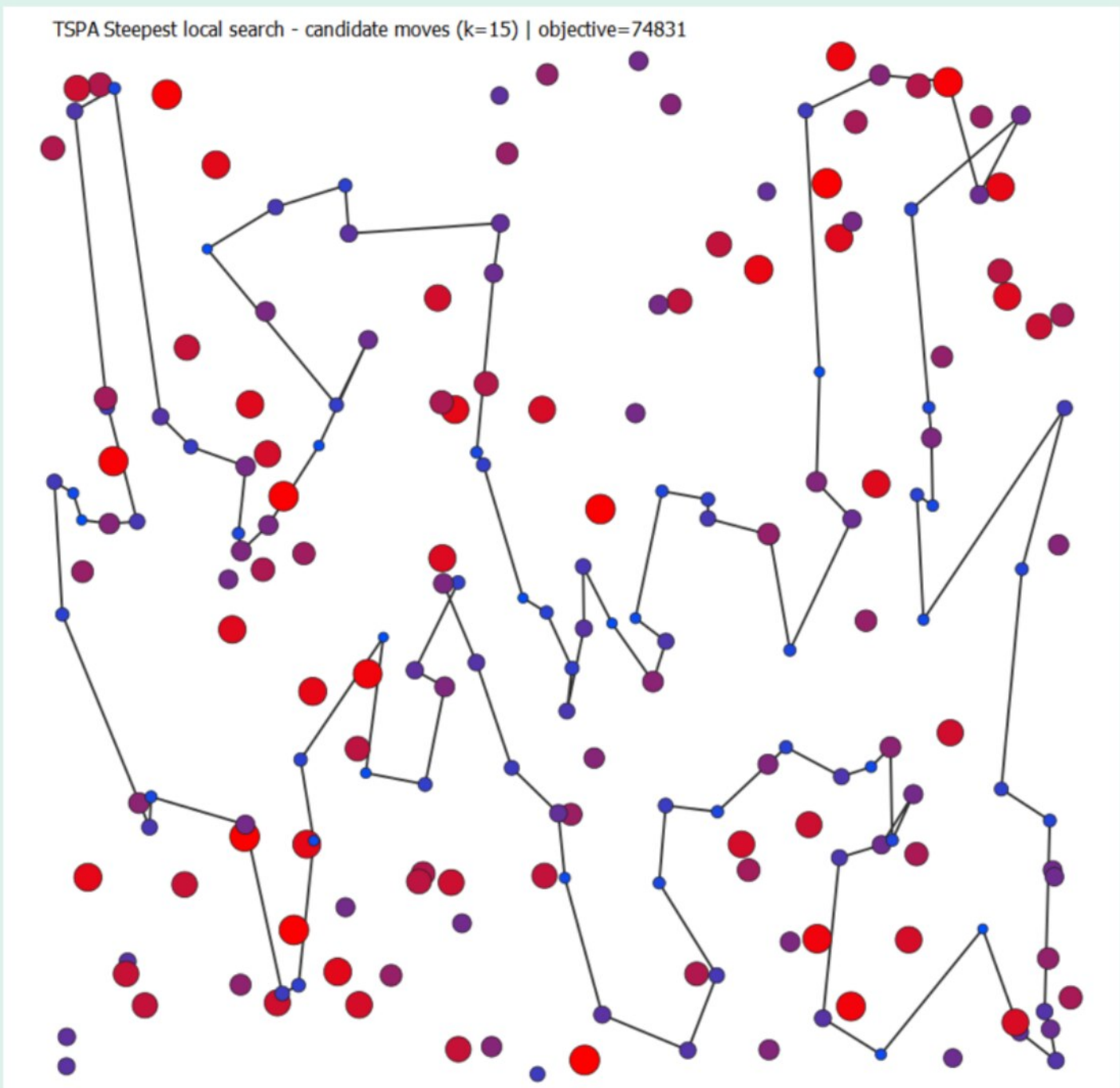
min=75168
max=81693
avg=77589.12

objective 75168

Best solution nodes:

182 138 151 42 89 106 150
176 72 134 84 58 158 192
181 135 114 25 70 98 87
141 30 187 2 171 92 104
129 69 127 97 95 168 63
149 116 60 49 108 152 100
123 132 76 38 80 48 91
157 71 4 16 51 35 180 28
102 185 67 144 45 73 140
53 52 179 177 131 36 33
143 9 54 174 8 164 136
184 178 29 55 101 59 1 3
169 50 93 47 147 130 159
170 153 193 78 88 107 57
182

100 selected nodes; closed route



BEST SOLUTION

Candidate moves (k = 15)

min=74831
max=81611
avg=77384.93

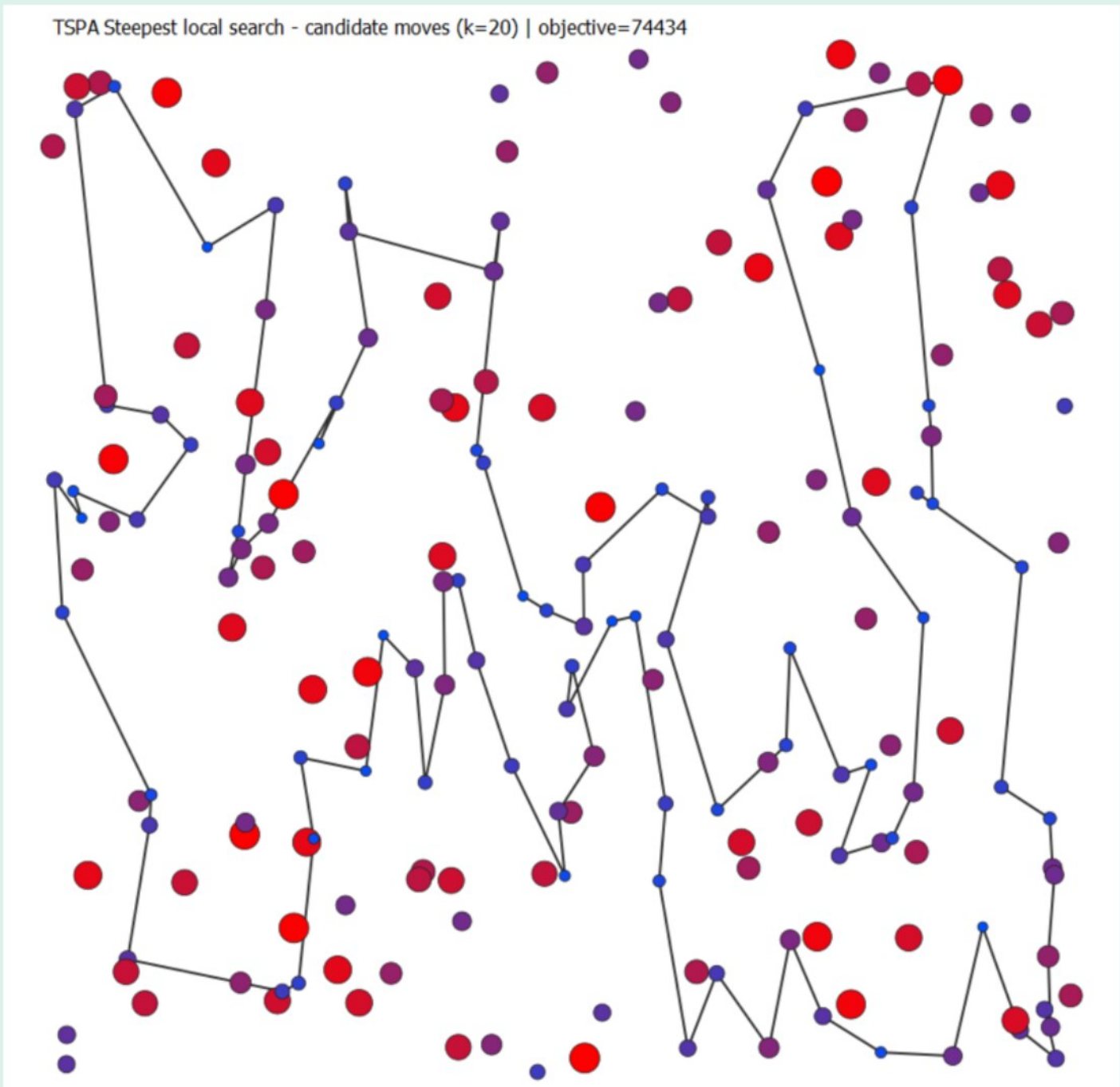
objective 74831

Best solution nodes:

33 36 131 99 177 52 179
140 53 144 67 16 51 139
112 71 157 91 80 76 4 132
90 83 123 100 152 108 48
31 60 82 49 11 63 149 116
168 127 97 161 171 2 187
30 180 141 87 102 28 35
156 185 45 73 135 181 192
158 58 84 134 72 176 150
106 43 89 113 42 151 182
57 107 88 78 193 153 159
130 47 147 93 3 50 169 1
178 55 101 29 163 136 164
142 85 174 8 54 9 33

100 selected nodes; closed route

8. Large best visualizations - TSPA



BEST SOLUTION

Candidate moves (k = 20)

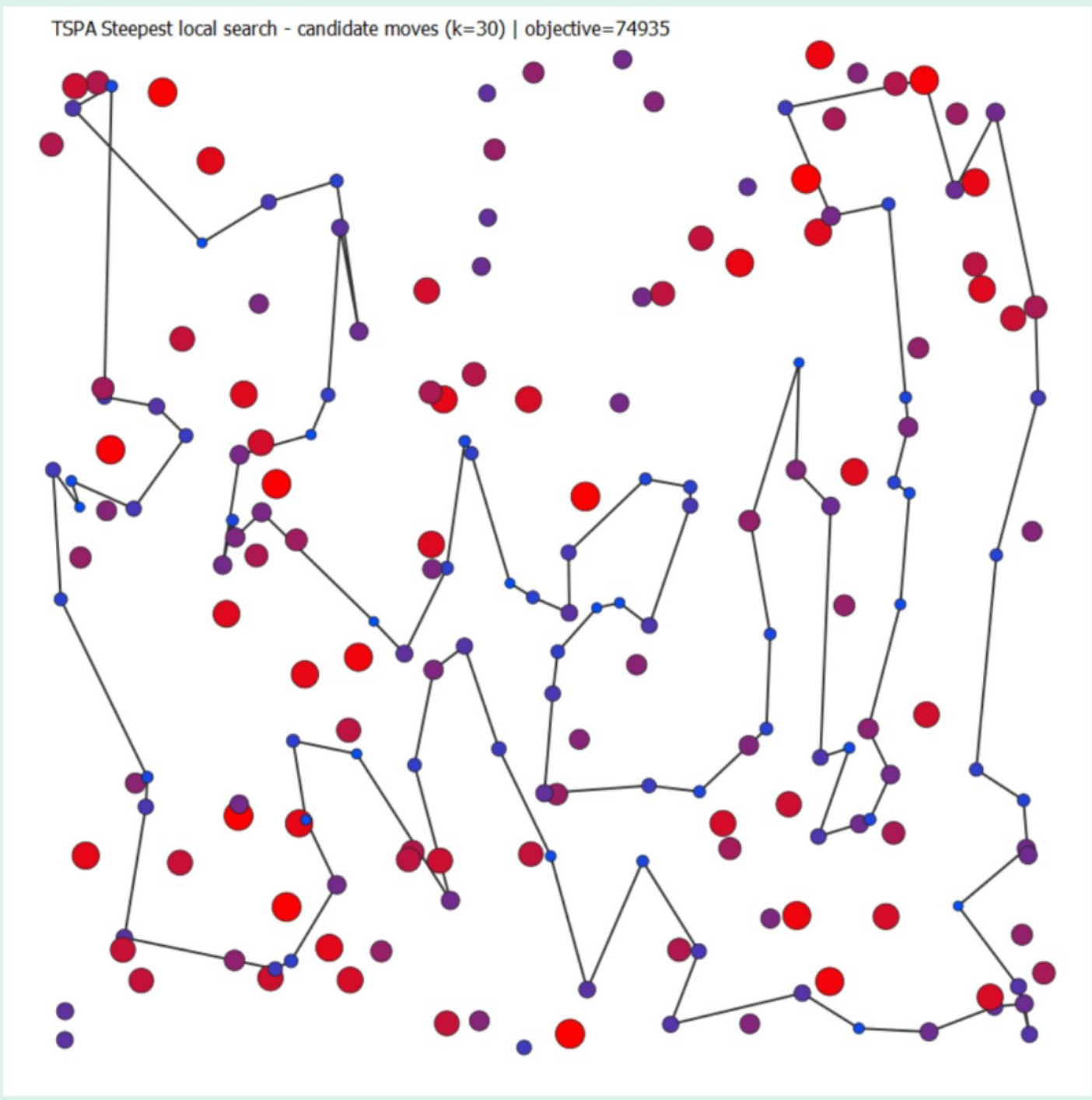
min=74434
max=80540
avg=77336.46

objective 74434

Best solution nodes:

176 72 134 131 9 54 33 52
179 144 67 16 51 112 139
71 157 4 132 76 90 83 160
123 100 38 91 80 48 31 82
60 108 152 49 149 63 116
168 97 127 69 171 2 187
30 141 180 28 87 102 156
35 185 45 135 73 41 53
140 177 36 84 58 192 158
111 138 182 57 64 107 88
78 153 193 75 170 146 159
130 47 3 50 169 1 178 29
136 186 164 85 93 113 89
42 151 106 150 174 176

100 selected nodes; closed route



BEST SOLUTION

Candidate moves (k = 30)

min=74935
max=80342
avg=77355.34

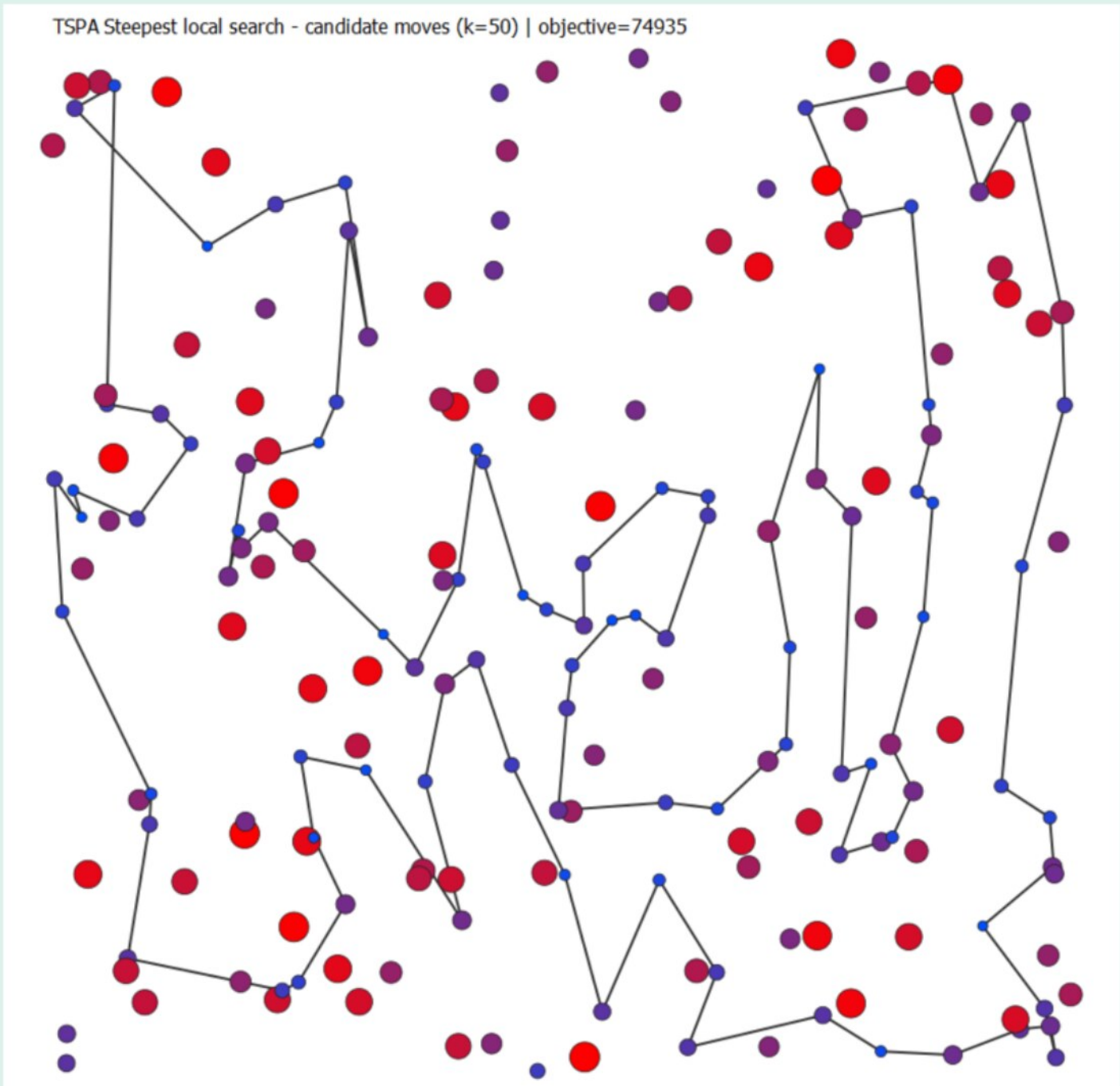
objective 74935

Best solution nodes:

134 84 73 140 53 177 36
131 54 9 33 52 179 144 67
16 51 35 28 180 90 83 123
160 100 132 76 71 4 157
91 80 31 48 60 108 152 49
149 63 116 168 97 127 69
171 2 189 187 30 141 98
87 102 185 45 135 181 58
158 192 182 57 64 88 193
78 153 107 146 170 159
130 47 147 21 55 101 29
136 184 178 1 169 3 50 93
43 113 89 42 151 106 150
85 142 164 8 174 176 134

100 selected nodes; closed route

8. Large best visualizations - TSPA



BEST SOLUTION

Candidate moves (k = 50)

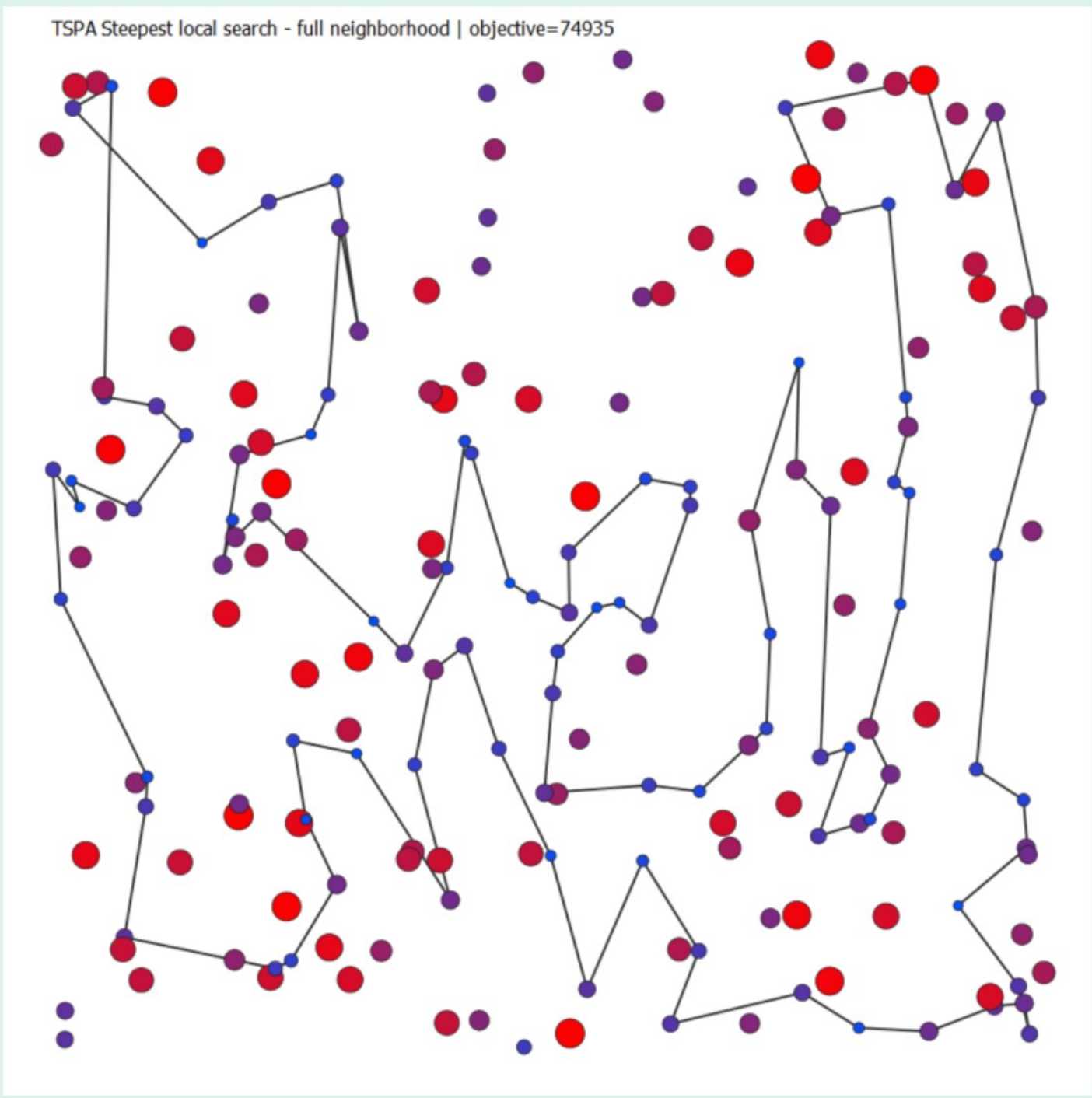
min=74935
max=80342
avg=77353.41

objective 74935

Best solution nodes:

134 84 73 140 53 177 36
131 54 9 33 52 179 144 67
16 51 35 28 180 90 83 123
160 100 132 76 71 4 157
91 80 31 48 60 108 152 49
149 63 116 168 97 127 69
171 2 189 187 30 141 98
87 102 185 45 135 181 58
158 192 182 57 64 88 193
78 153 107 146 170 159
130 47 147 21 55 101 29
136 184 178 1 169 3 50 93
43 113 89 42 151 106 150
85 142 164 8 174 176 134

100 selected nodes; closed route



BEST SOLUTION

Full neighborhood (k = infinity)

min=74935
max=80342
avg=77354.36

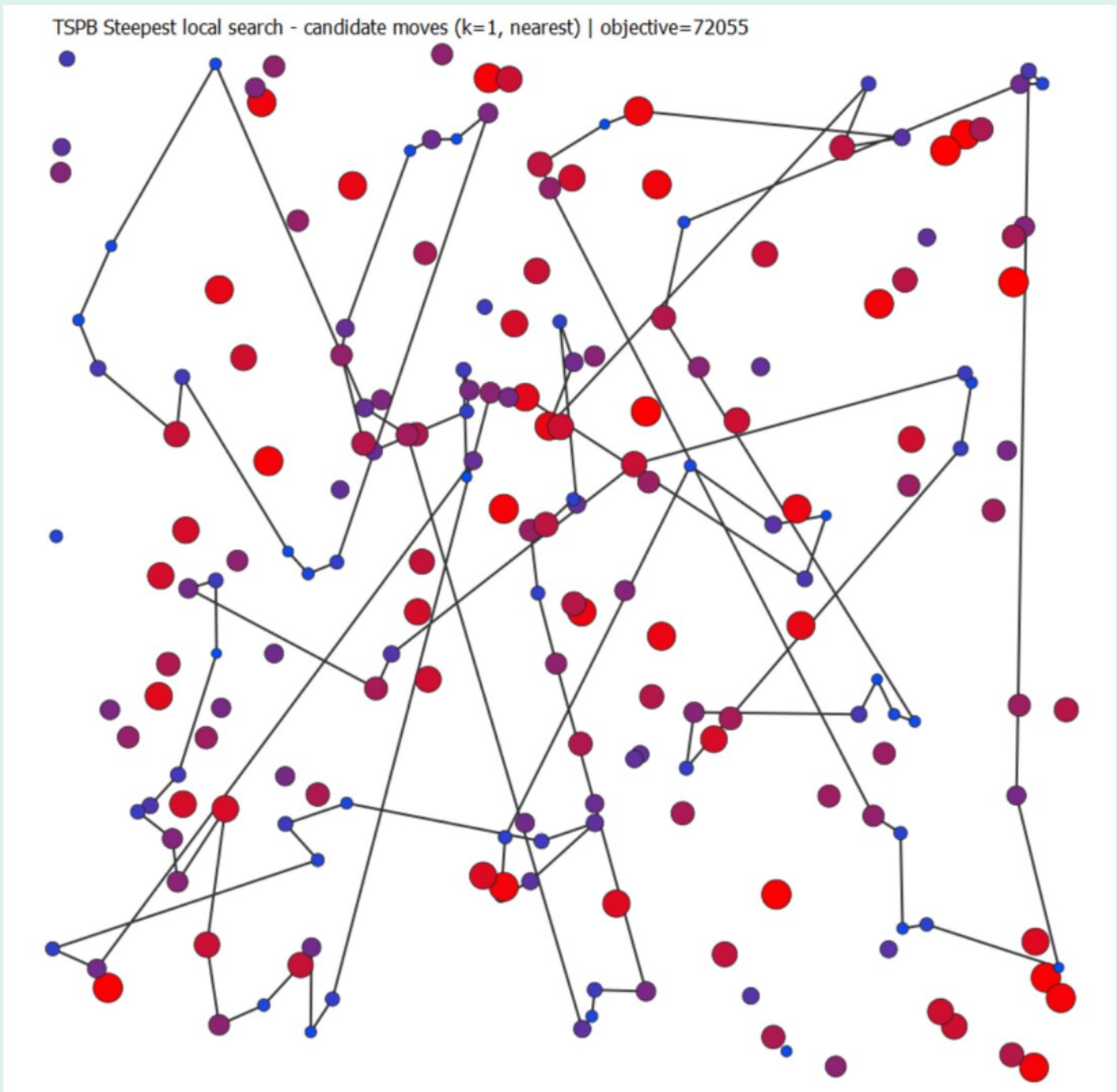
objective 74935

Best solution nodes:

134 84 73 140 53 177 36
131 54 9 33 52 179 144 67
16 51 35 28 180 90 83 123
160 100 132 76 71 4 157
91 80 31 48 60 108 152 49
149 63 116 168 97 127 69
171 2 189 187 30 141 98
87 102 185 45 135 181 58
158 192 182 57 64 88 193
78 153 107 146 170 159
130 47 147 21 55 101 29
136 184 178 1 169 3 50 93
43 113 89 42 151 106 150
85 142 164 8 174 176 134

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Candidate moves (k = 1, nearest)

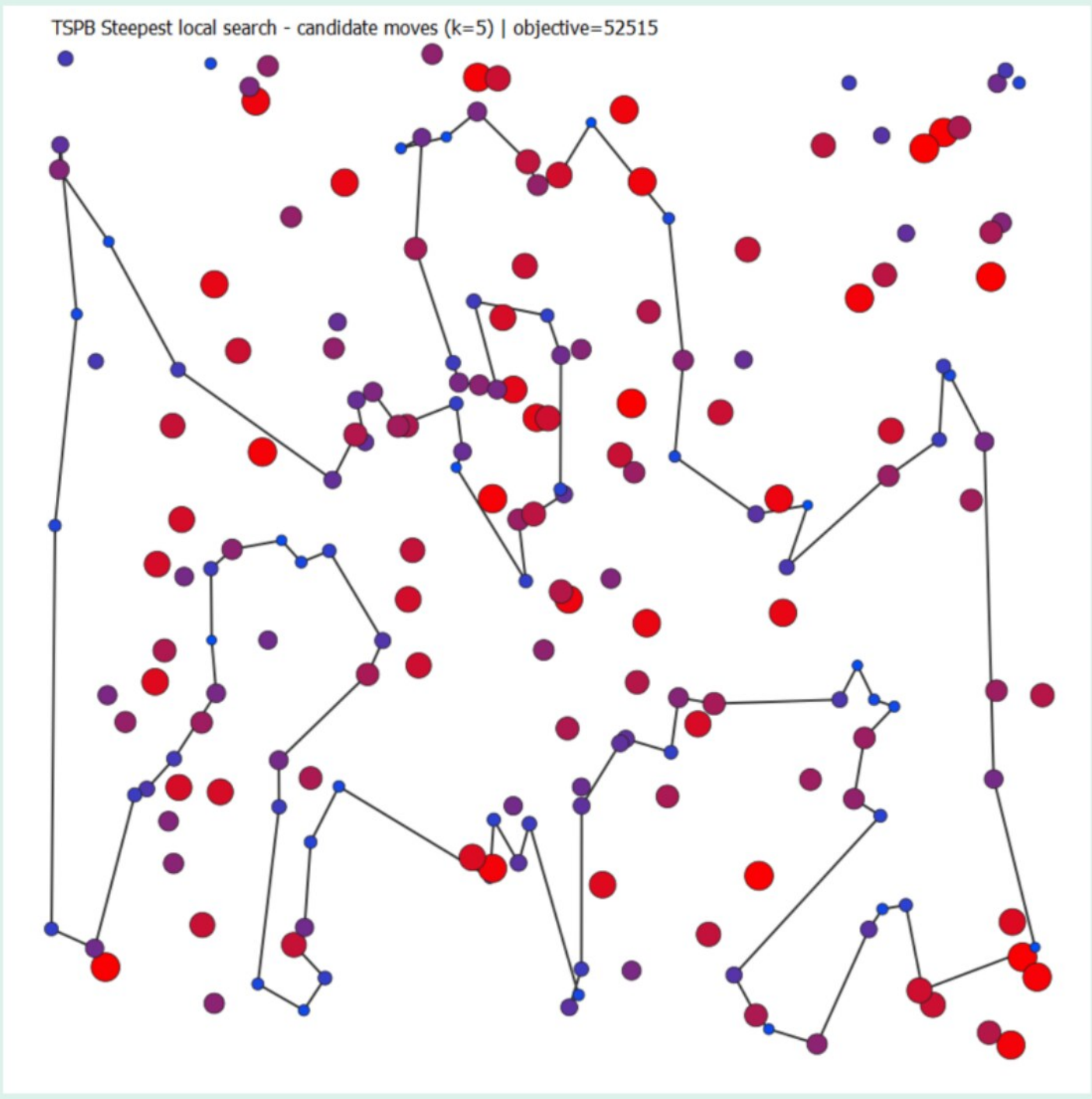
min=72055
max=90620
avg=81385.45

objective 72055

Best solution nodes:

174 160 69 133 136 172 21
176 191 130 121 198 96 48
47 140 154 158 138 145 28
149 116 30 189 82 87 179
88 98 151 122 173 184 3
72 46 109 190 194 70 44
89 43 156 182 153 20 14
110 118 94 105 81 57 79
90 129 127 80 66 67 10
114 196 75 64 33 177 125
170 6 29 132 197 36 113
91 62 175 142 157 134 45
8 117 95 192 54 162 76 59
85 51 181 167 40 71 60 1
174

100 selected nodes; closed route



BEST SOLUTION

Candidate moves (k = 5)

min=52515
max=58105
avg=54893.31

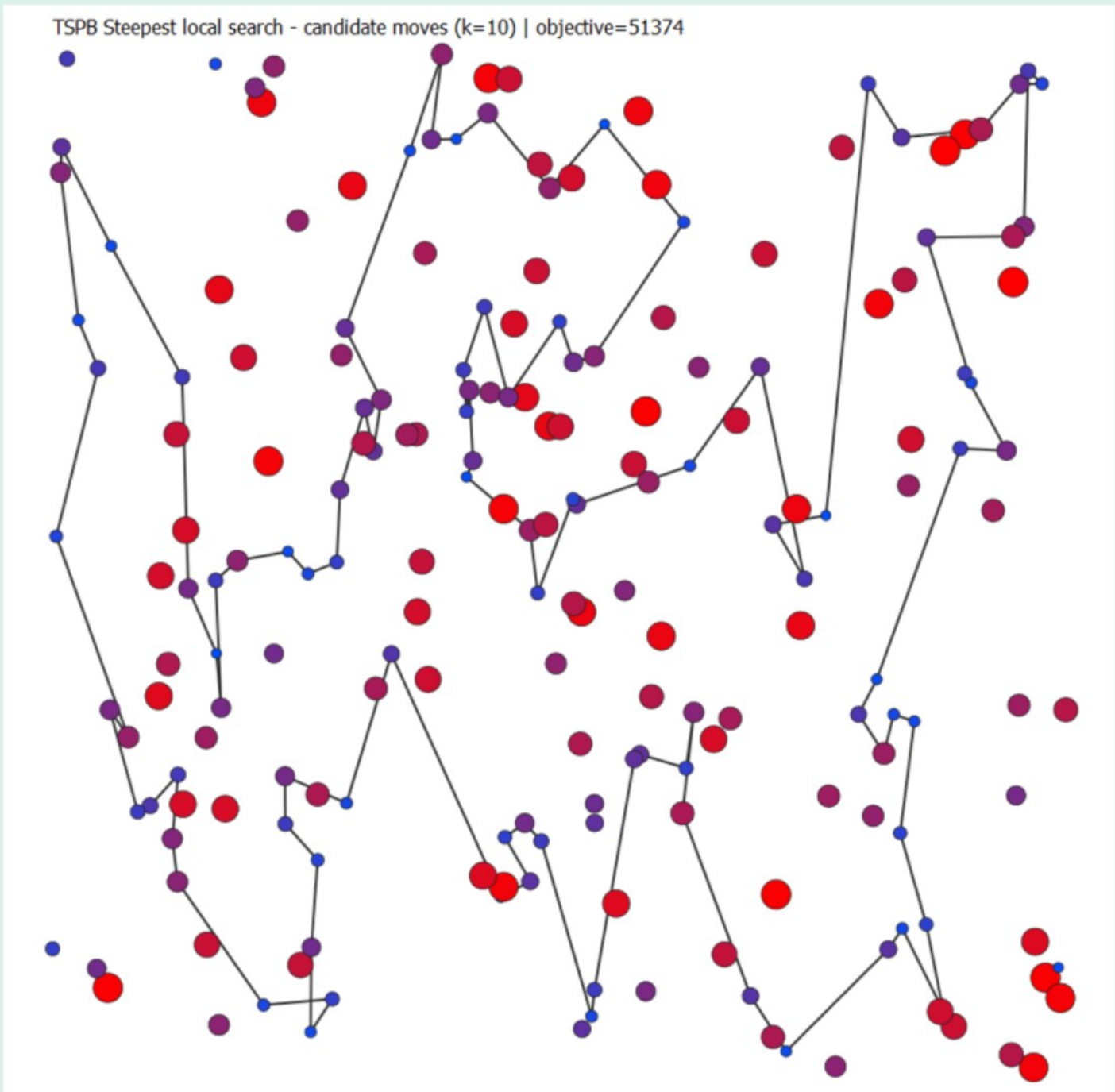
objective 52515

Best solution nodes:

155 71 175 134 45 142 157
60 40 85 59 51 167 87 82
179 181 34 24 67 66 84 80
127 129 90 52 182 153 99
39 171 17 74 20 14 2 110
118 148 114 196 10 78 192
54 162 76 185 57 89 32
156 43 48 96 121 198 63
133 136 8 117 120 72 46
184 3 173 122 151 69 160
172 189 183 30 21 176 38
158 149 13 146 28 65 1
174 132 29 6 115 170 125
26 154 140 47 64 33 155

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Candidate moves (k = 10)

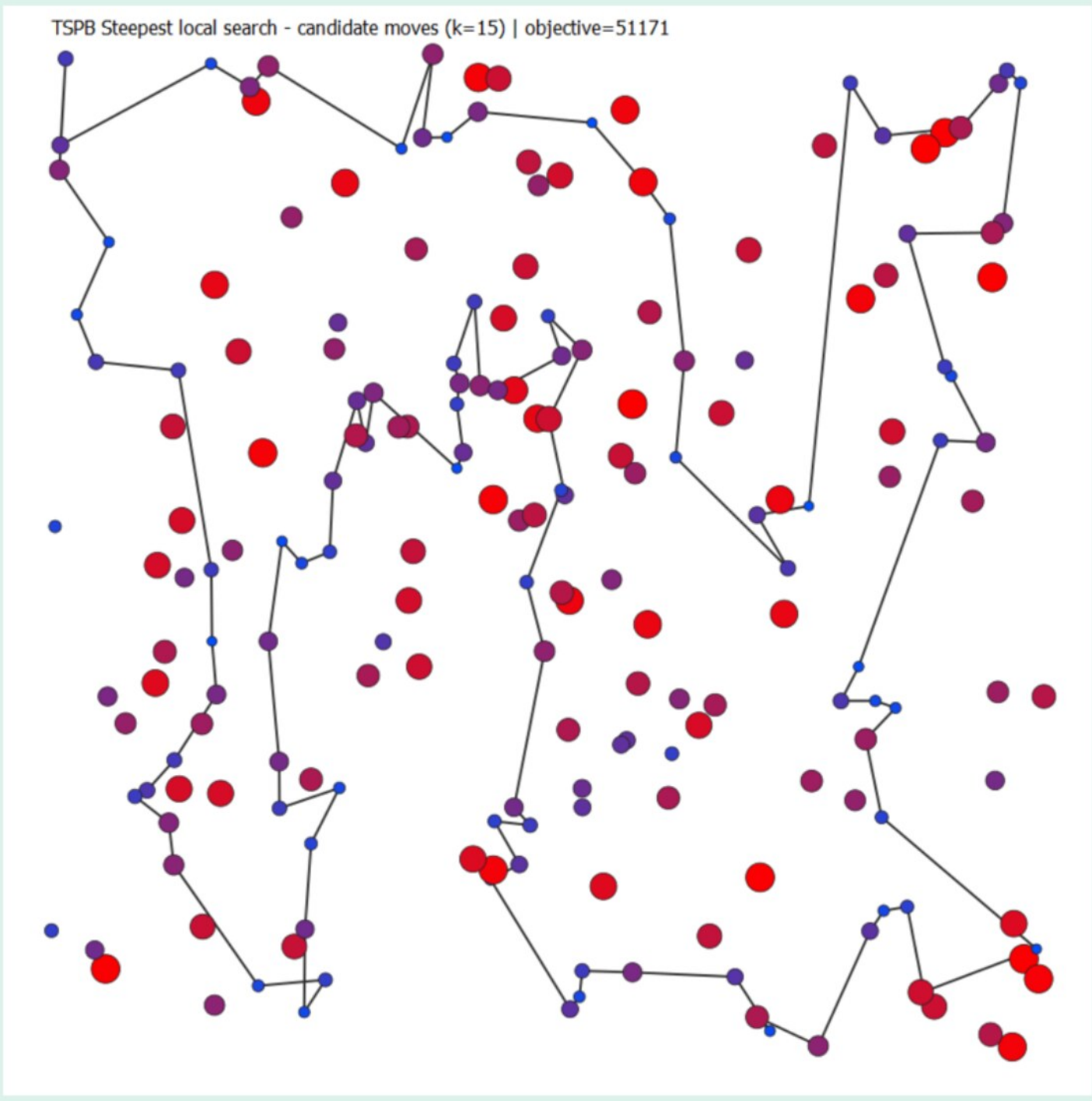
min=51374
max=56520
avg=53573.29

objective 51374

Best solution nodes:

```
70 190 54 162 192 104 76
3 184 151 122 69 160 136
172 133 120 117 72 46 86
57 89 156 48 96 198 58
121 130 183 21 30 38 47
140 154 26 125 115 170
177 158 149 146 13 28 145
65 103 178 132 29 6 197
36 175 45 134 157 60 71
155 40 64 85 51 59 73 167
87 179 34 24 67 66 31 99
171 74 20 2 14 153 90 129
52 80 127 10 148 196 114
187 139 61 94 105 81 169
70
```

100 selected nodes; closed route



BEST SOLUTION

Candidate moves (k = 15)

min=51171
max=56408
avg=53472.66

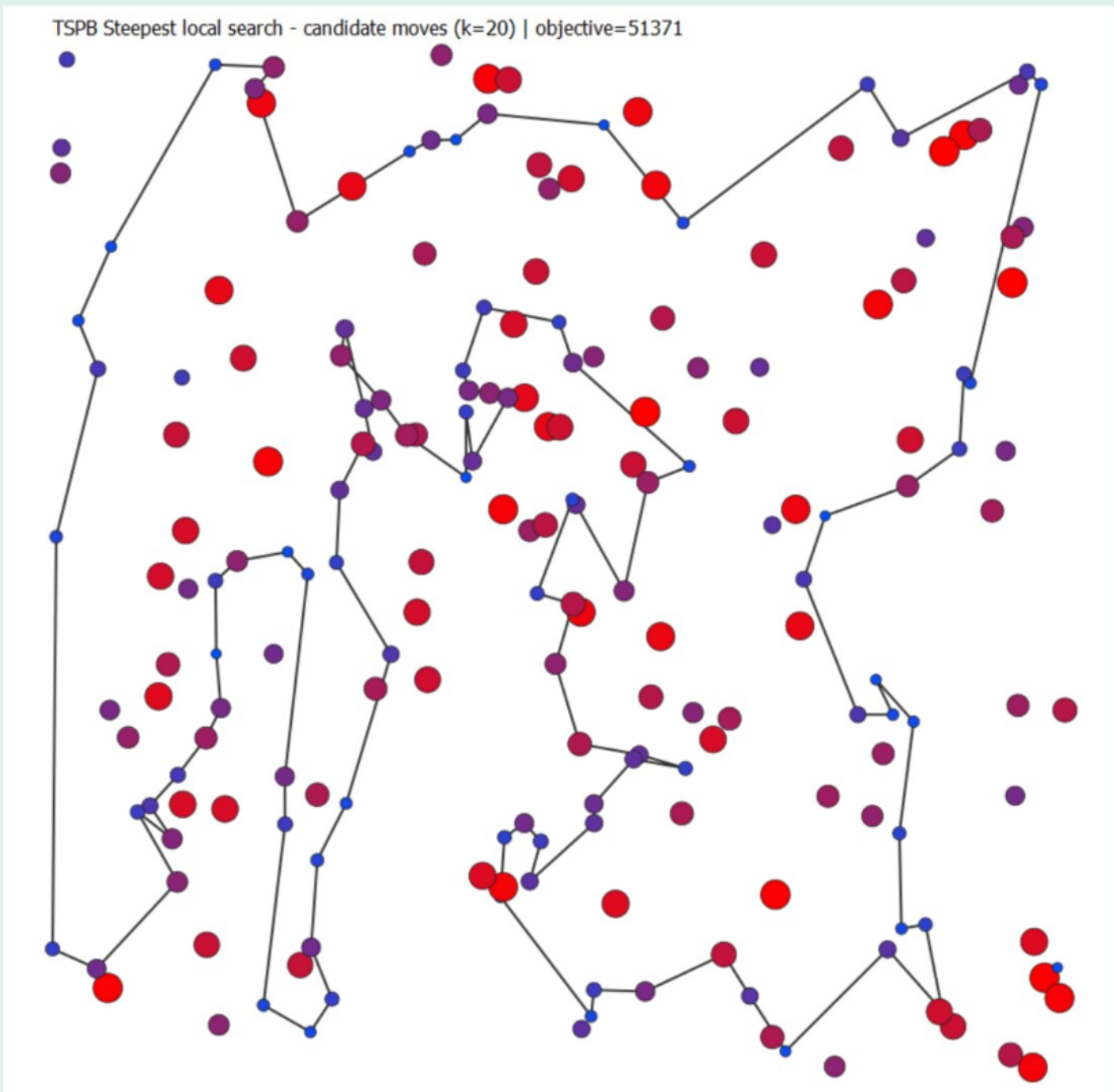
objective 51171

Best solution nodes:

```
190 70 169 81 94 105 61
139 187 196 114 148 10
127 80 129 90 52 153 110
2 14 20 74 17 39 171 99
88 179 87 82 85 51 59 167
73 98 151 184 3 137 86 72
46 117 8 120 133 136 172
160 69 183 21 30 38 47
140 154 97 155 71 40 60
157 134 45 175 36 197 132
29 6 115 170 125 158 145
28 149 13 37 146 116 147
42 121 58 198 96 48 89 83
57 185 76 192 162 54 190
```

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Candidate moves (k = 20)

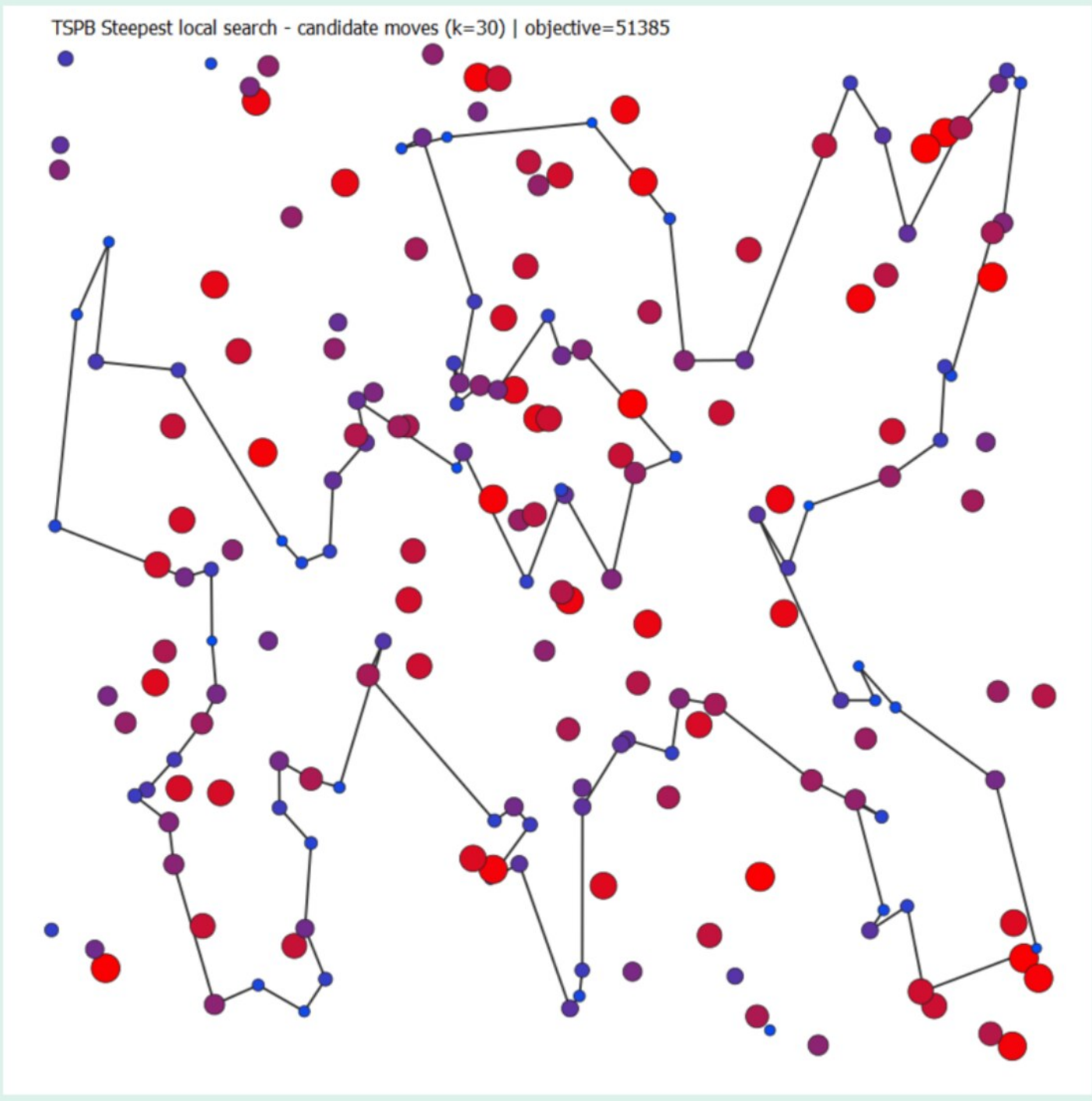
min=51371
max=56469
avg=53461.22

objective 51371

Best solution nodes:

180 115 170 125 26 154
140 155 71 175 134 45 157
60 40 64 47 38 176 21 30
130 191 183 189 69 172
160 117 8 136 133 120 72
46 76 135 27 3 184 151
111 98 25 24 67 34 152
181 51 167 73 59 85 87
179 88 163 99 171 74 2 14
20 153 90 127 129 80 192
54 78 10 196 114 105 94
70 190 57 89 48 96 198
121 50 147 42 116 149 28
145 65 1 174 36 132 197
29 6 180

100 selected nodes; closed route



BEST SOLUTION

Candidate moves (k = 30)

min=51385
max=56469
avg=53515.31

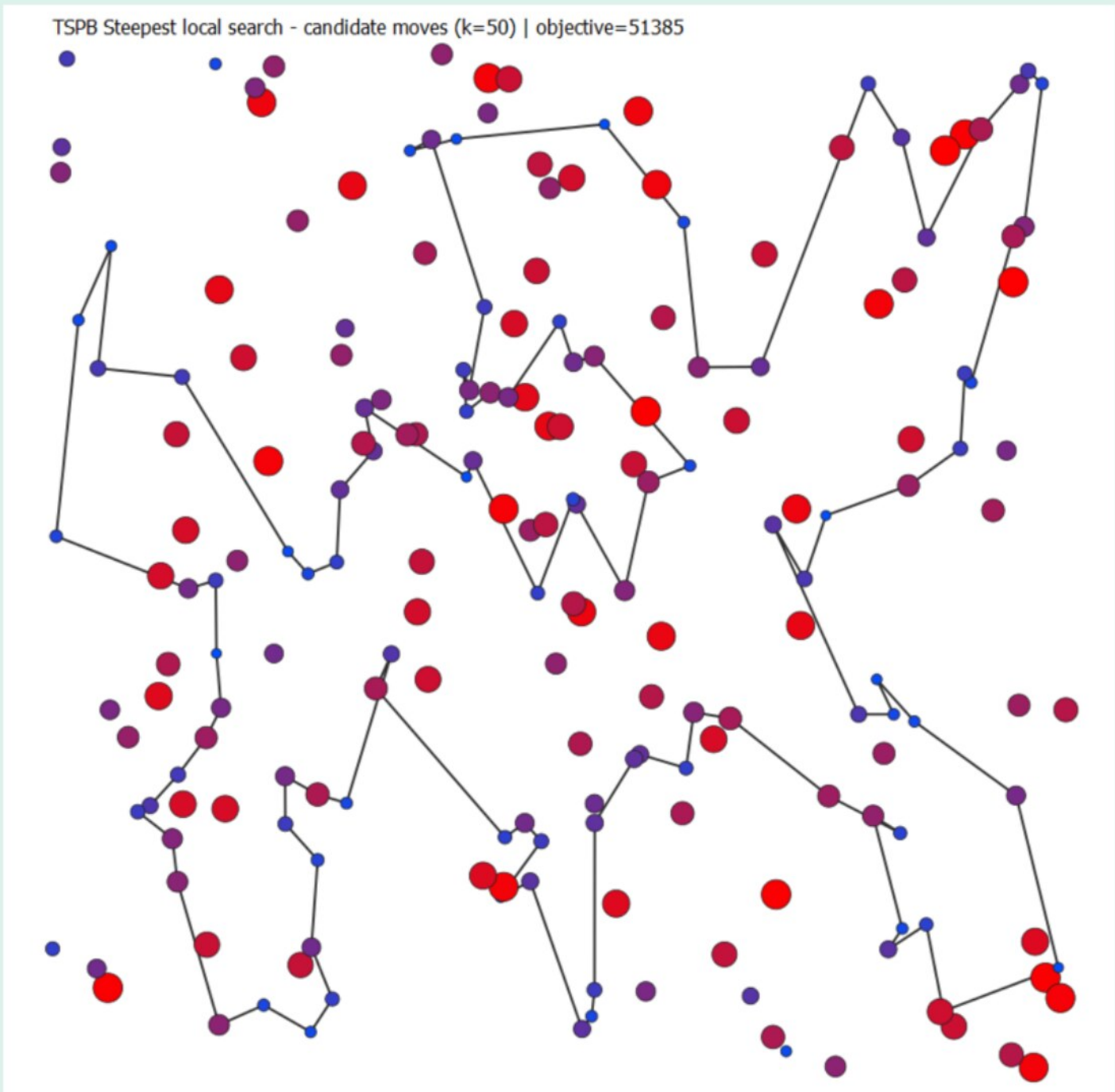
objective 51385

Best solution nodes:

96 121 198 120 136 133
172 8 117 72 46 86 76 135
27 3 184 151 160 69 30 21
38 47 140 154 158 145 149
28 65 177 125 170 115 180
6 29 132 197 36 62 175
134 45 157 60 71 155 141
40 64 33 59 73 167 85 51
82 87 179 181 34 24 67 66
84 150 153 182 20 74 14 2
110 118 90 127 129 80 162
192 54 78 10 196 114 139
61 105 94 81 169 187 70
190 104 185 57 89 96

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Candidate moves (k = 50)

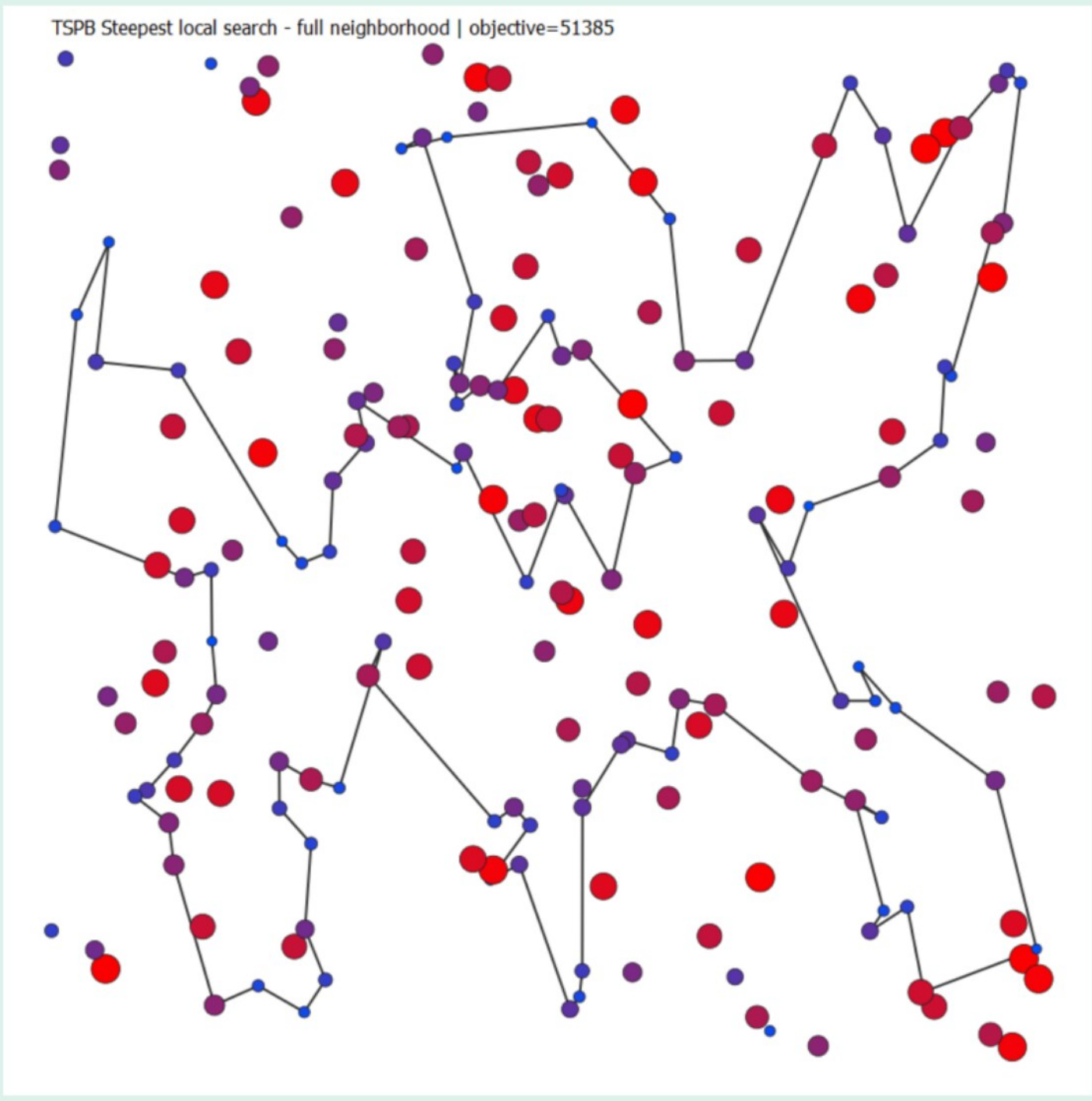
min=51385
max=56469
avg=53506.79

objective 51385

Best solution nodes:

```
96 121 198 120 136 133
172 8 117 72 46 86 76 135
27 3 184 151 160 69 30 21
38 47 140 154 158 145 149
28 65 177 125 170 115 180
6 29 132 197 36 62 175
134 45 157 60 71 155 141
40 64 33 59 73 167 85 51
82 87 179 181 34 24 67 66
84 150 153 182 20 74 14 2
110 118 90 127 129 80 162
192 54 78 10 196 114 139
61 105 94 81 169 187 70
190 104 185 57 89 96
```

100 selected nodes; closed route



BEST SOLUTION

Full neighborhood (k = infinity)

min=51385
max=56469
avg=53505.13

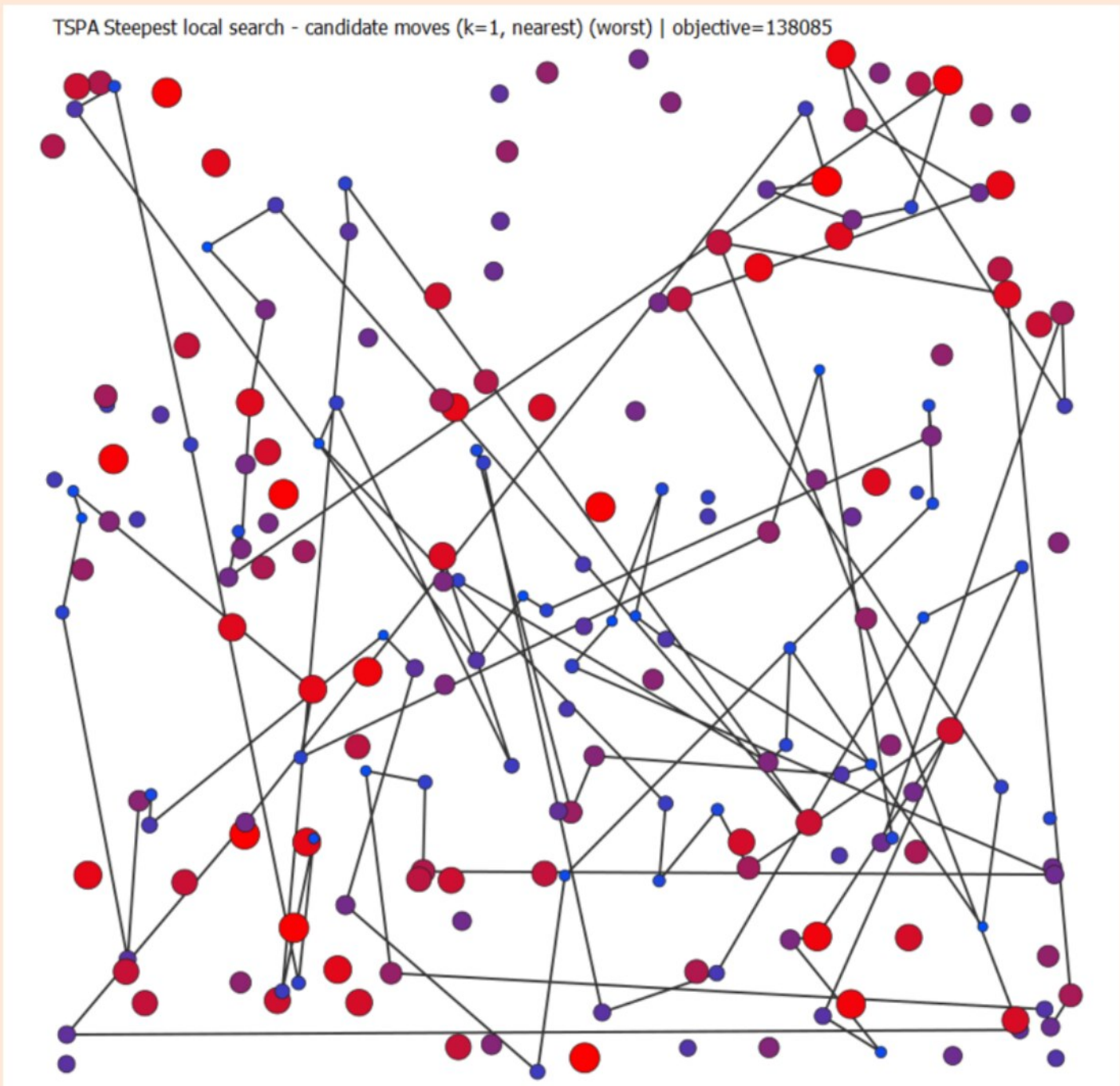
objective 51385

Best solution nodes:

```
96 121 198 120 136 133
172 8 117 72 46 86 76 135
27 3 184 151 160 69 30 21
38 47 140 154 158 145 149
28 65 177 125 170 115 180
6 29 132 197 36 62 175
134 45 157 60 71 155 141
40 64 33 59 73 167 85 51
82 87 179 181 34 24 67 66
84 150 153 182 20 74 14 2
110 118 90 127 129 80 162
192 54 78 10 196 114 139
61 105 94 81 169 187 70
190 104 185 57 89 96
```

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Candidate moves (k = 1, nearest)

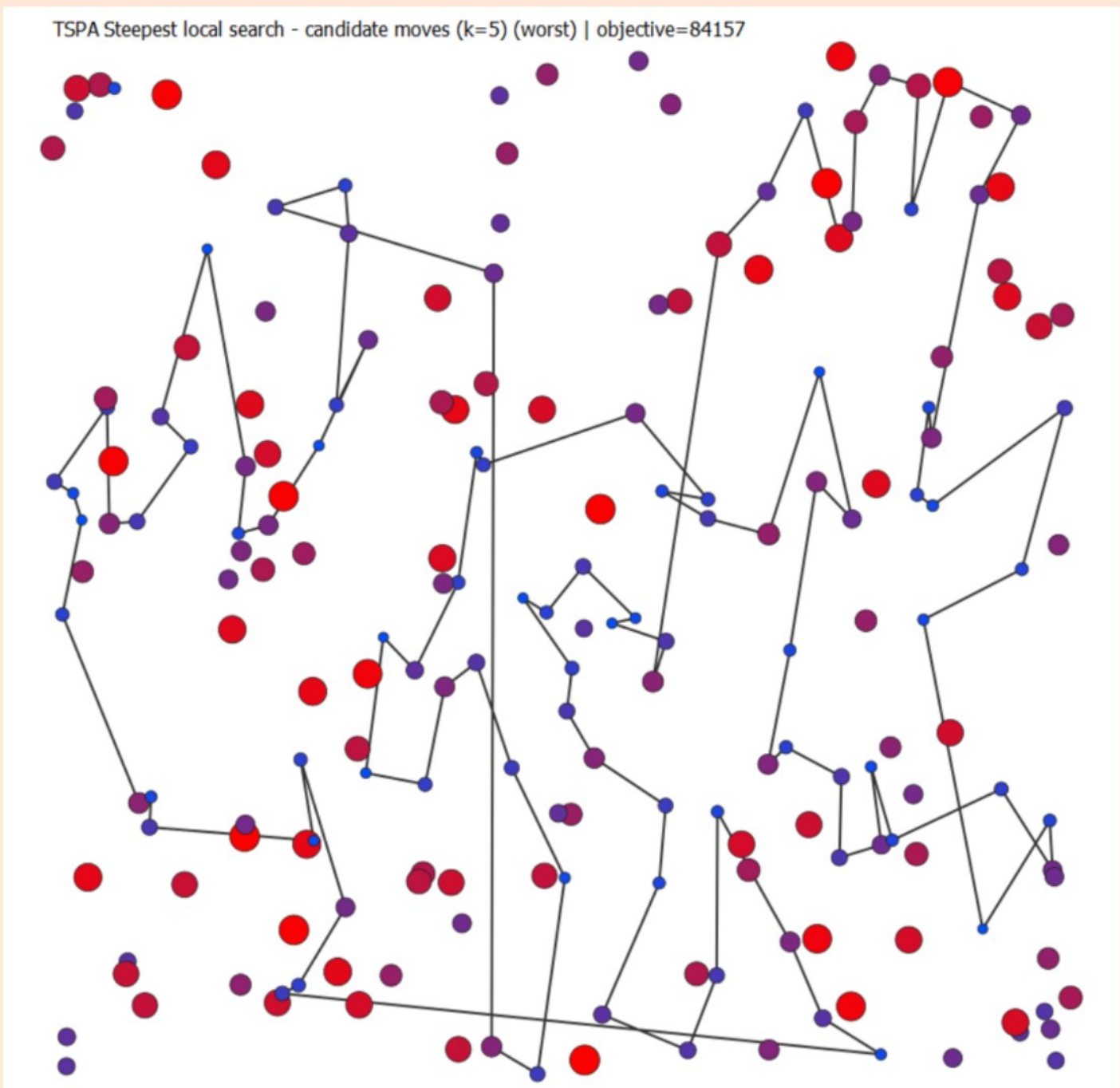
min=108540
max=138085
avg=119836.08

objective 138085

Worst solution nodes:

```
101 24 197 147 21 42 89
164 8 30 23 149 63 168 69
95 97 127 180 28 189 114
135 50 1 169 144 67 185
31 48 152 2 187 171 71
157 0 91 80 38 14 83 123
160 29 178 184 186 195
136 161 129 88 34 155 26
193 153 10 141 87 27 170
53 177 33 36 131 106 150
41 61 51 16 181 158 93 47
182 57 138 165 117 65 134
58 84 132 76 45 172 156
35 72 176 174 107 130 15
101
```

100 selected nodes; closed route



WORST SOLUTION

Candidate moves (k = 5)

min=74889
max=84157
avg=79905.04

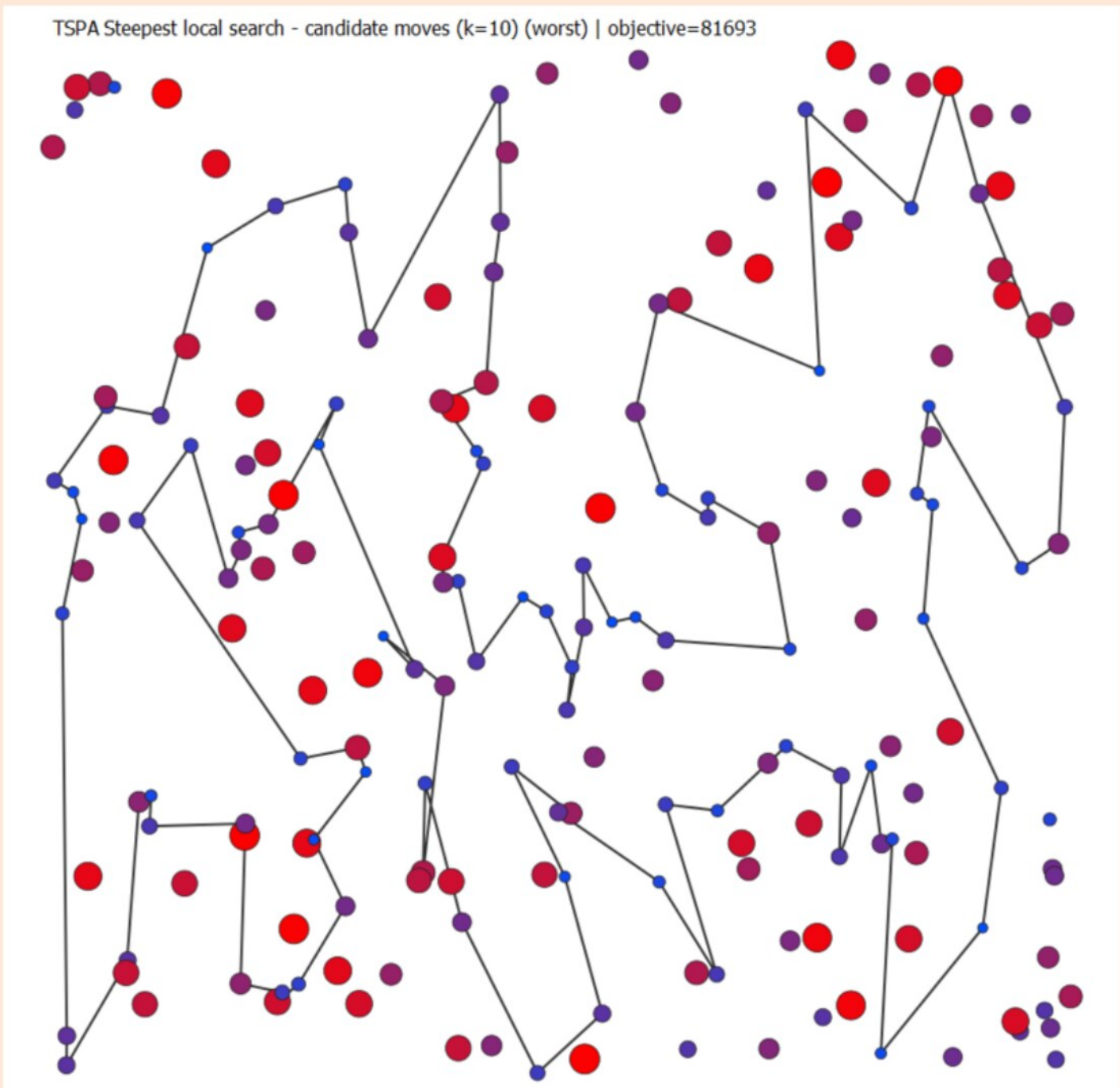
objective 84157

Worst solution nodes:

```
163 19 178 29 55 101 169
1 3 50 147 47 93 107 159
146 130 89 106 42 151 150
176 72 174 142 85 164 194
8 54 33 9 143 16 51 35 28
180 141 87 102 185 45 135
114 25 139 91 157 71 76 4
132 90 123 100 80 108 152
49 11 60 116 149 63 168
95 97 127 187 30 189 2
171 57 182 138 134 158
192 181 58 84 41 140 53
67 144 52 36 177 131 99
34 186 136 56 184 24 163
```

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Candidate moves (k = 10)

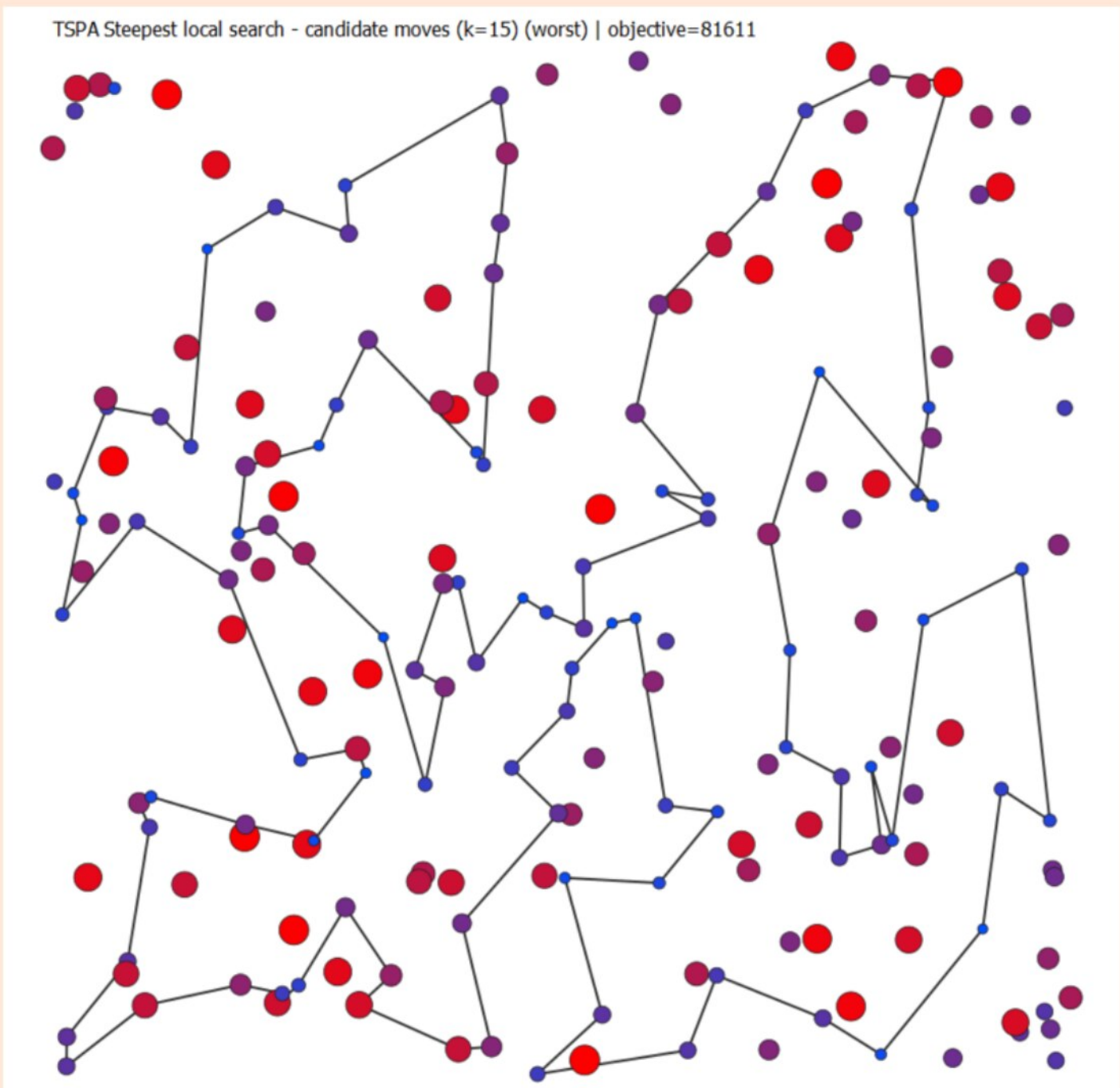
min=75168
max=81693
avg=77589.12

objective 81693

Worst solution nodes:

93 130 107 57 89 42 106
151 150 176 134 84 158 58
73 61 45 135 181 114 98
87 27 102 180 28 132 76
90 123 83 160 152 49 30
145 141 187 189 2 32 171
92 161 127 97 95 94 69
104 129 168 63 149 116 60
108 80 91 157 71 4 115
112 139 133 196 51 16 172
156 35 185 67 144 53 140
179 52 177 36 131 174 8 9
54 33 143 62 164 136 178
29 101 147 37 47 1 3 50
93

100 selected nodes; closed route



WORST SOLUTION

Candidate moves (k = 15)

min=74831
max=81611
avg=77384.93

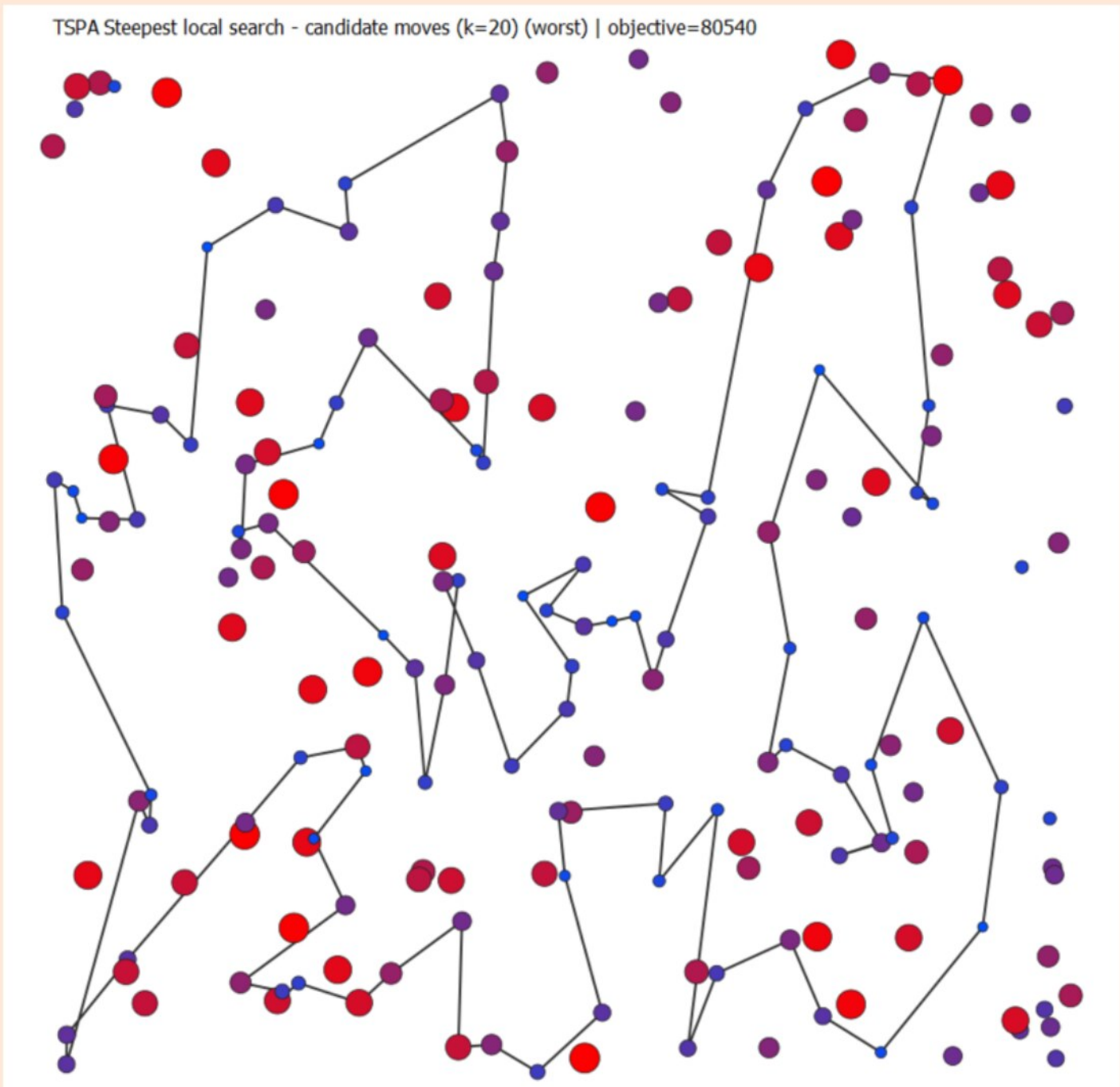
objective 81611

Worst solution nodes:

157 71 91 80 152 108 60
149 63 168 49 160 30 145
141 187 161 97 95 127 69
129 104 110 92 171 32 2
189 10 183 22 25 98 73 45
140 53 177 36 84 134 58
135 181 114 192 158 182
57 107 130 159 47 93 89
106 42 151 150 176 174 8
164 50 3 1 178 29 163 136
186 62 143 9 33 54 52 179
144 67 185 35 156 28 102
87 180 90 123 100 132 76
4 51 16 139 112 5 115 157

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Candidate moves (k = 20)

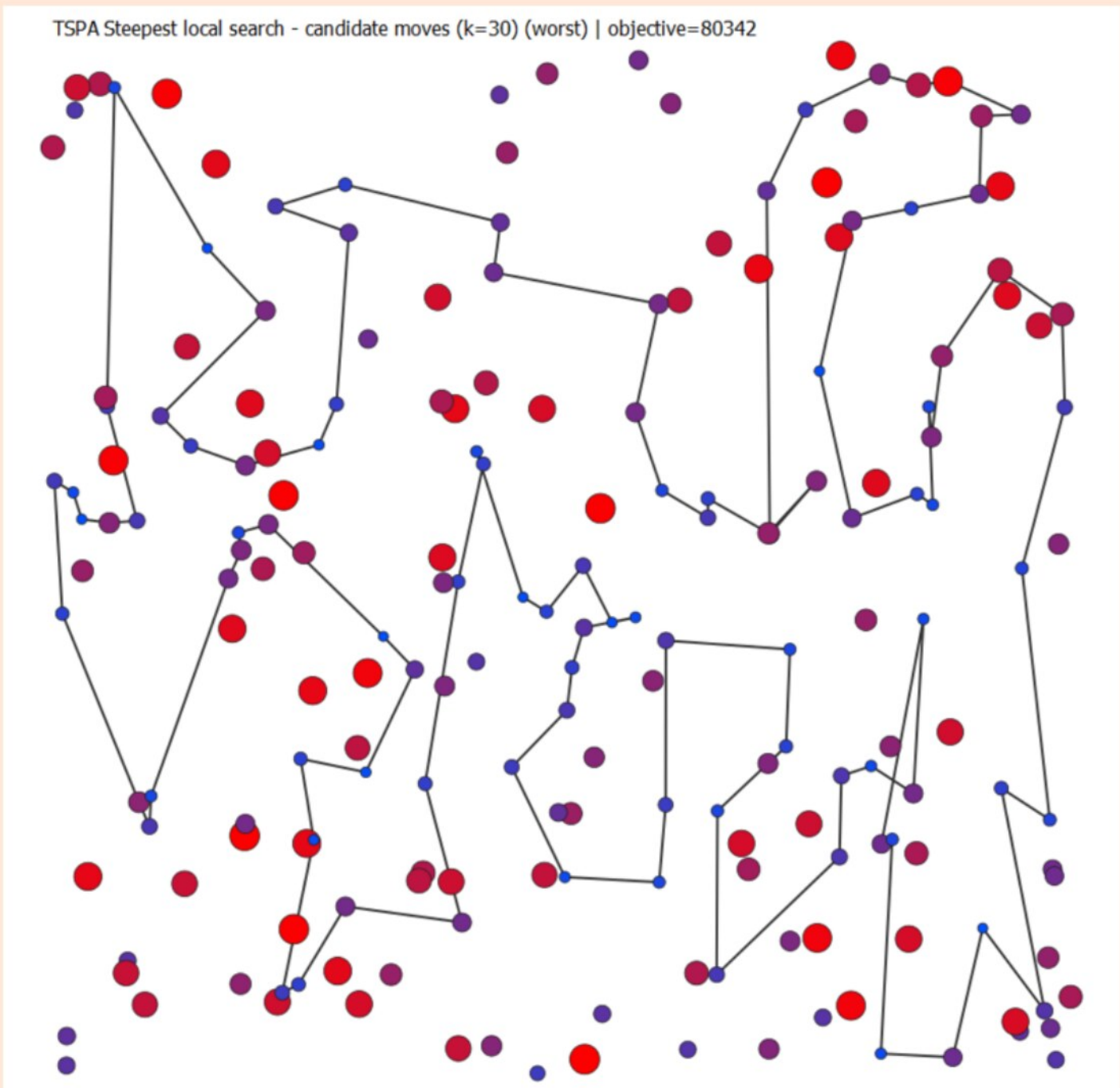
min=74434
max=80540
avg=77336.46

objective 80540

Worst solution nodes:

157 71 91 80 152 108 60
49 63 149 116 168 97 127
95 104 129 69 161 30 145
141 187 189 92 171 32 2
183 10 98 22 25 114 181
135 73 84 58 134 192 158
138 182 57 107 130 93 106
89 151 42 150 176 72 174
8 164 50 3 1 178 29 163
136 186 9 33 54 131 99 36
177 179 144 52 67 53 140
45 185 156 35 102 87 28
180 90 123 83 100 132 76
4 51 16 139 112 5 115 157

100 selected nodes; closed route



WORST SOLUTION

Candidate moves (k = 30)

min=74935
max=80342
avg=77355.34

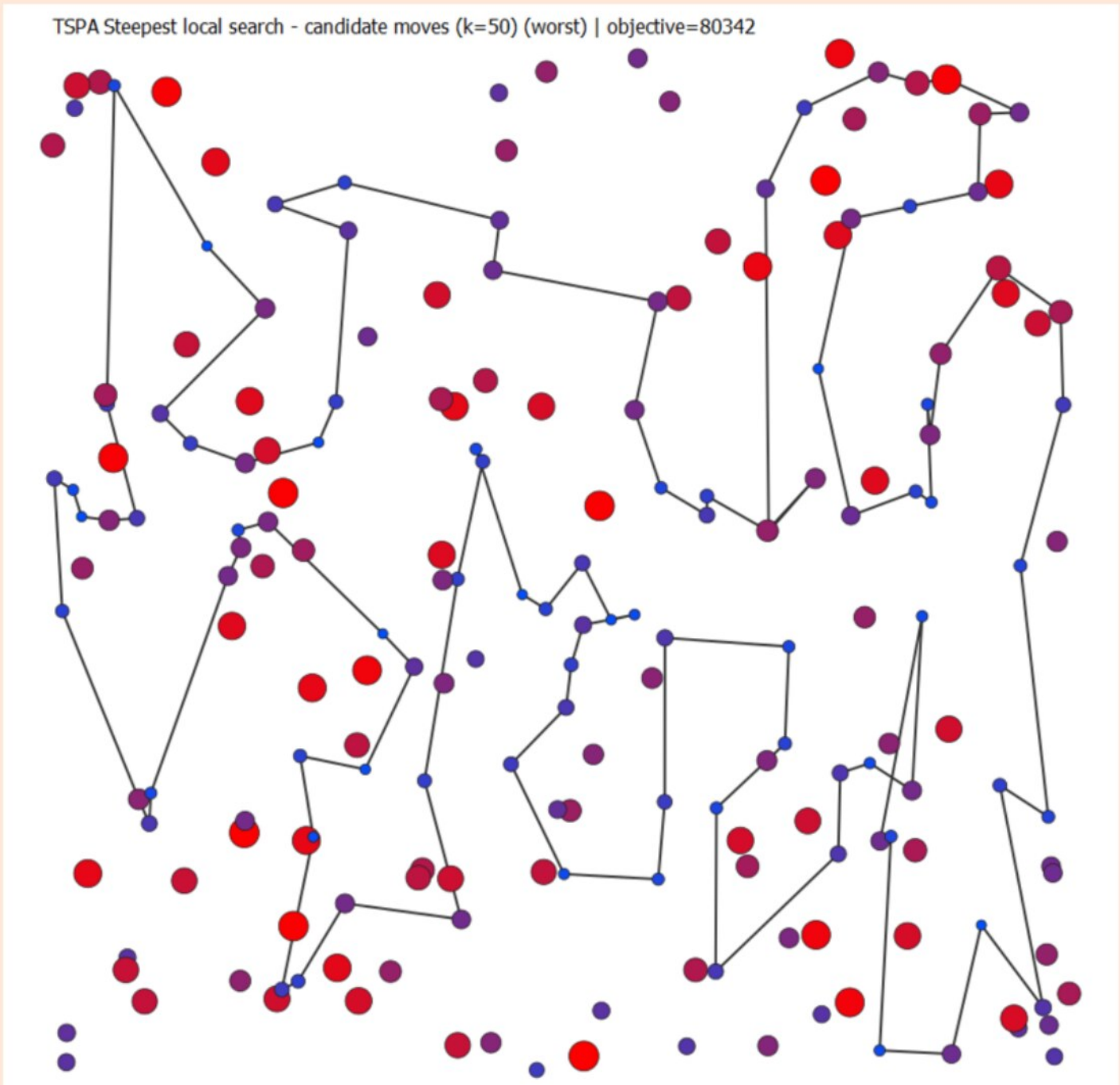
objective 80342

Worst solution nodes:

123 83 160 97 127 95 168
116 149 63 11 49 60 48 80
38 108 152 100 132 76 71
91 157 112 139 62 143 33
54 9 194 142 8 186 136
163 19 29 55 17 101 178
184 164 85 3 50 1 169 59
46 21 147 47 159 130 153
107 64 57 89 42 93 113
106 150 151 158 134 72
176 174 131 84 58 135 45
140 53 179 36 177 52 144
67 51 16 35 87 98 189 2
171 187 30 141 28 180 90
123

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Candidate moves (k = 50)

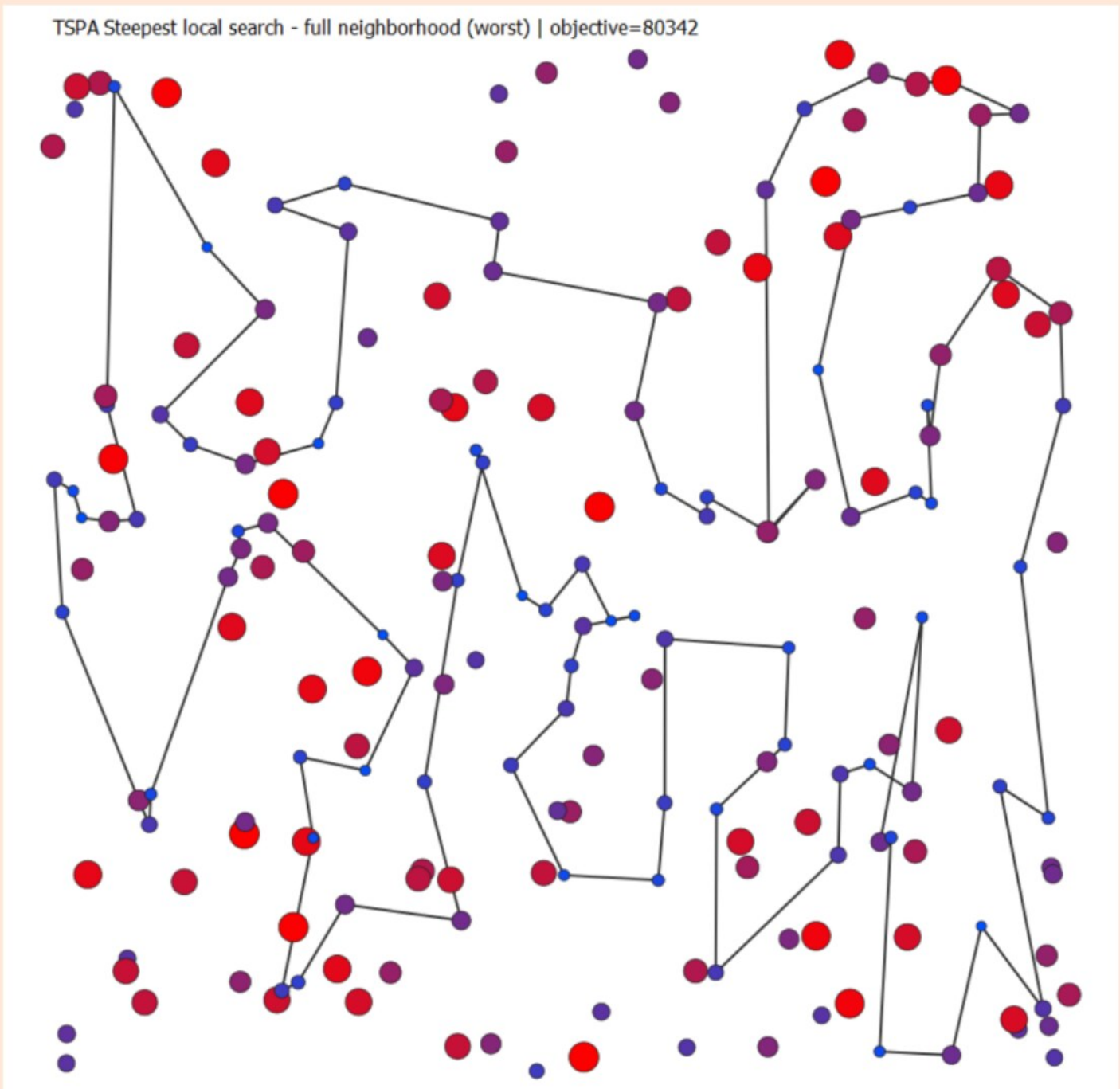
min=74935
max=80342
avg=77353.41

objective 80342

Worst solution nodes:

123 83 160 97 127 95 168
116 149 63 11 49 60 48 80
38 108 152 100 132 76 71
91 157 112 139 62 143 33
54 9 194 142 8 186 136
163 19 29 55 17 101 178
184 164 85 3 50 1 169 59
46 21 147 47 159 130 153
107 64 57 89 42 93 113
106 150 151 158 134 72
176 174 131 84 58 135 45
140 53 179 36 177 52 144
67 51 16 35 87 98 189 2
171 187 30 141 28 180 90
123

100 selected nodes; closed route



WORST SOLUTION

Full neighborhood (k = infinity)

min=74935
max=80342
avg=77354.36

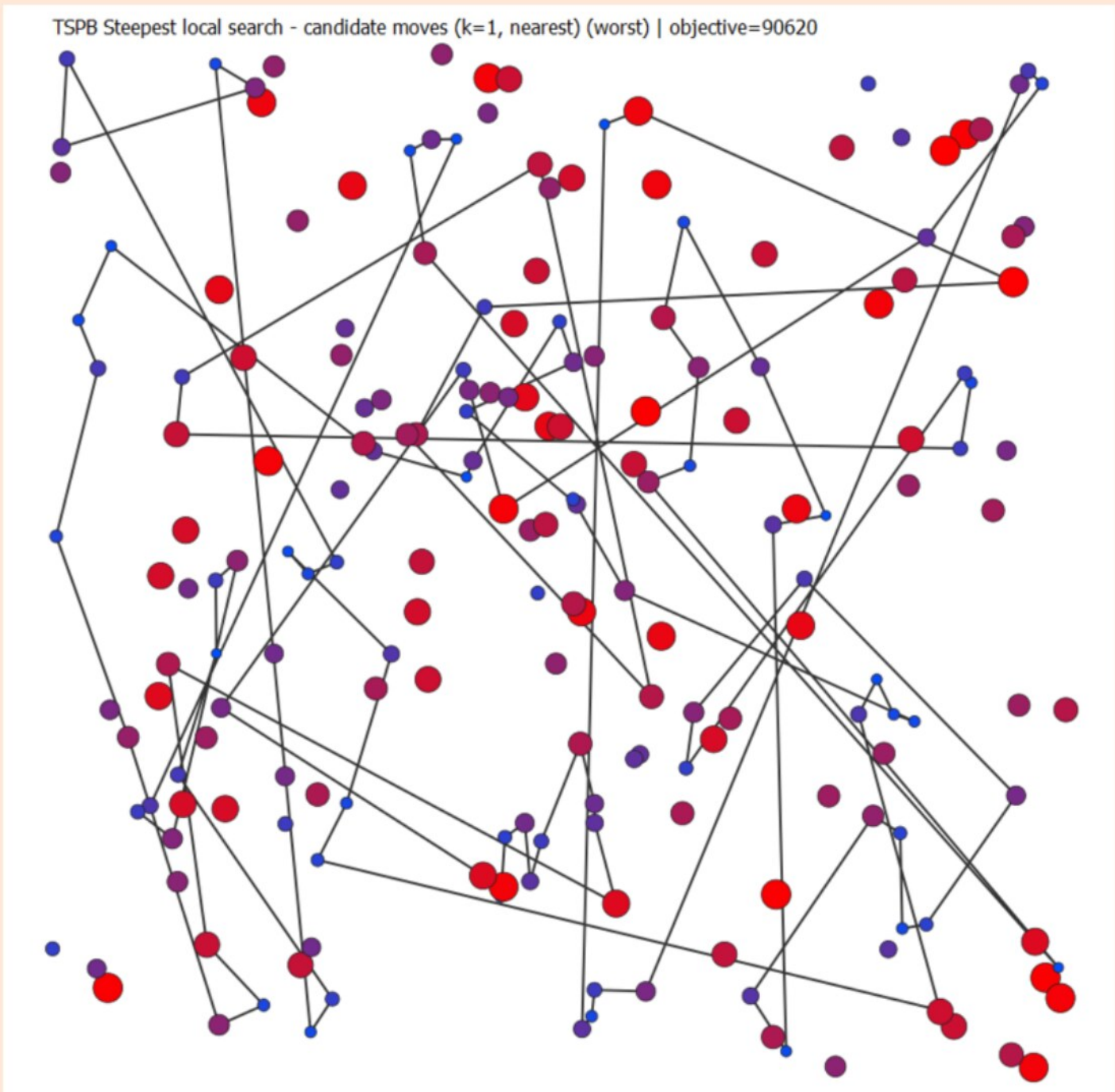
objective 80342

Worst solution nodes:

123 83 160 97 127 95 168
116 149 63 11 49 60 48 80
38 108 152 100 132 76 71
91 157 112 139 62 143 33
54 9 194 142 8 186 136
163 19 29 55 17 101 178
184 164 85 3 50 1 169 59
46 21 147 47 159 130 153
107 64 57 89 42 93 113
106 150 151 158 134 72
176 174 131 84 58 135 45
140 53 179 36 177 52 144
67 51 16 35 87 98 189 2
171 187 30 141 28 180 90
123

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Candidate moves (k = 1, nearest)

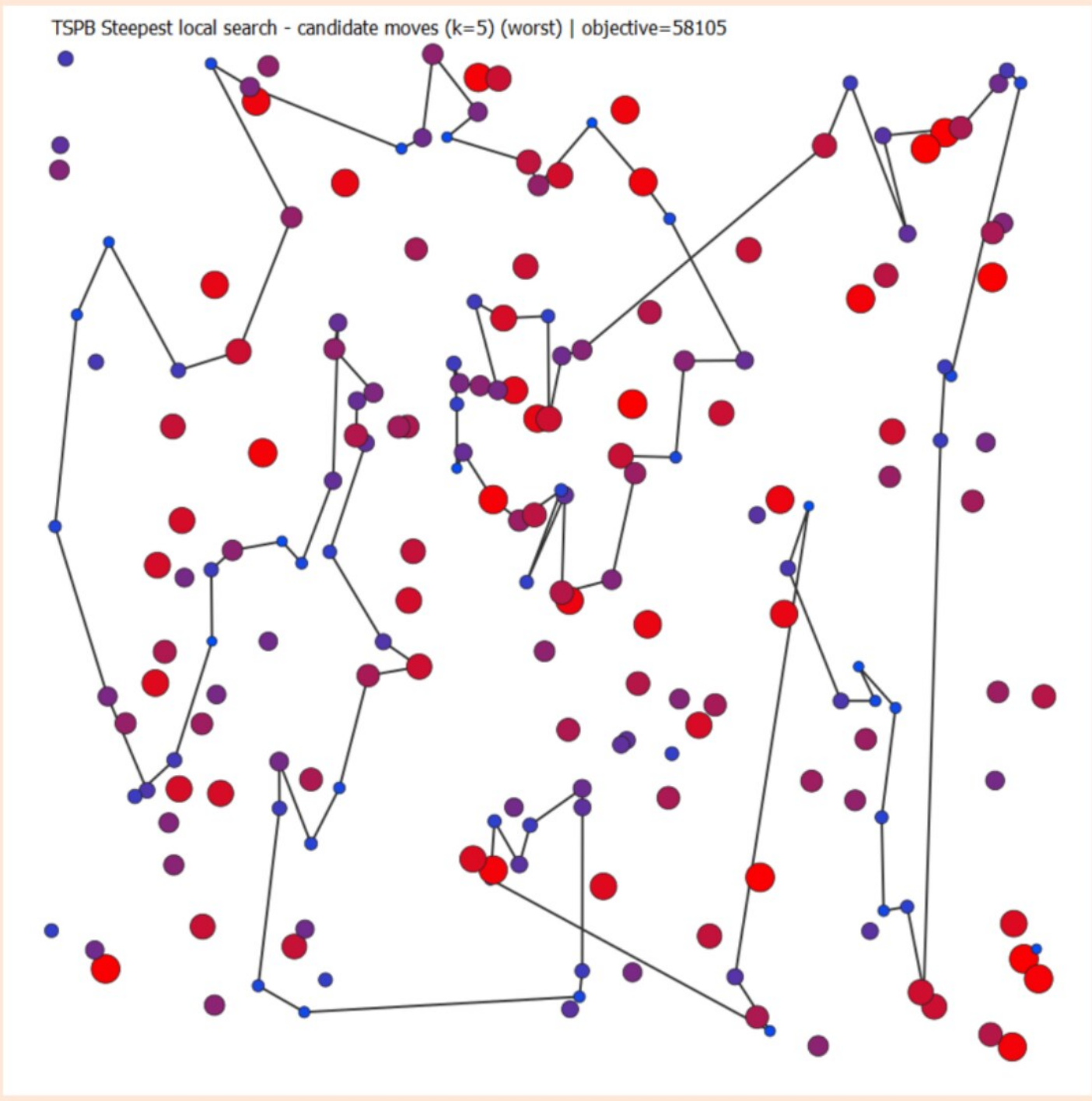
min=72055
max=90620
avg=81385.45

objective 90620

Worst solution nodes:

```
82 89 44 193 120 168 189
11 43 158 138 10 114 196
67 66 192 118 14 20 153
182 99 171 162 54 104 57
79 185 76 135 110 68 63
121 198 96 29 132 197 26
125 170 6 45 134 116 147
146 37 47 140 154 64 40
60 2 188 80 127 129 90 27
184 3 172 46 72 160 69 21
149 28 145 65 103 62 175
91 19 126 181 152 25 167
51 73 59 85 107 115 133
128 187 105 94 88 179 87
82
```

100 selected nodes; closed route



WORST SOLUTION

Candidate moves (k = 5)

min=52515
max=58105
avg=54893.31

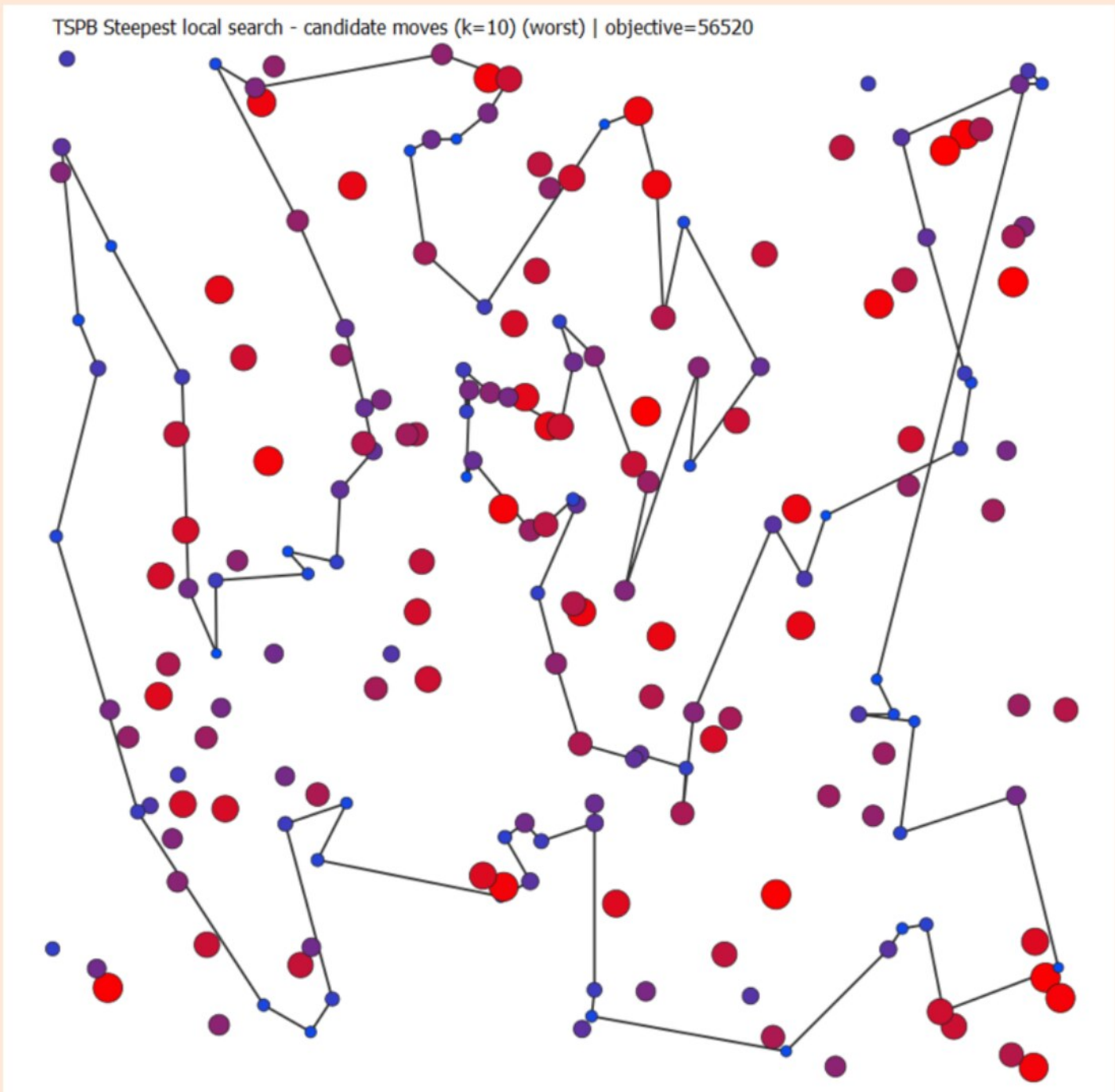
objective 58105

Worst solution nodes:

```
147 121 198 58 48 96 43
156 89 83 57 104 185 76
75 135 27 111 3 151 184
122 128 160 69 172 133
136 8 117 120 92 72 137
46 86 194 190 187 70 169
81 94 105 114 196 10 2 14
20 153 90 127 129 80 192
54 99 171 85 59 51 167
152 181 179 87 134 175 71
155 60 40 33 49 64 47 21
176 30 183 191 130 38 140
154 26 125 170 6 132 29
178 65 28 149 158 15 50
116 147
```

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Candidate moves (k = 10)

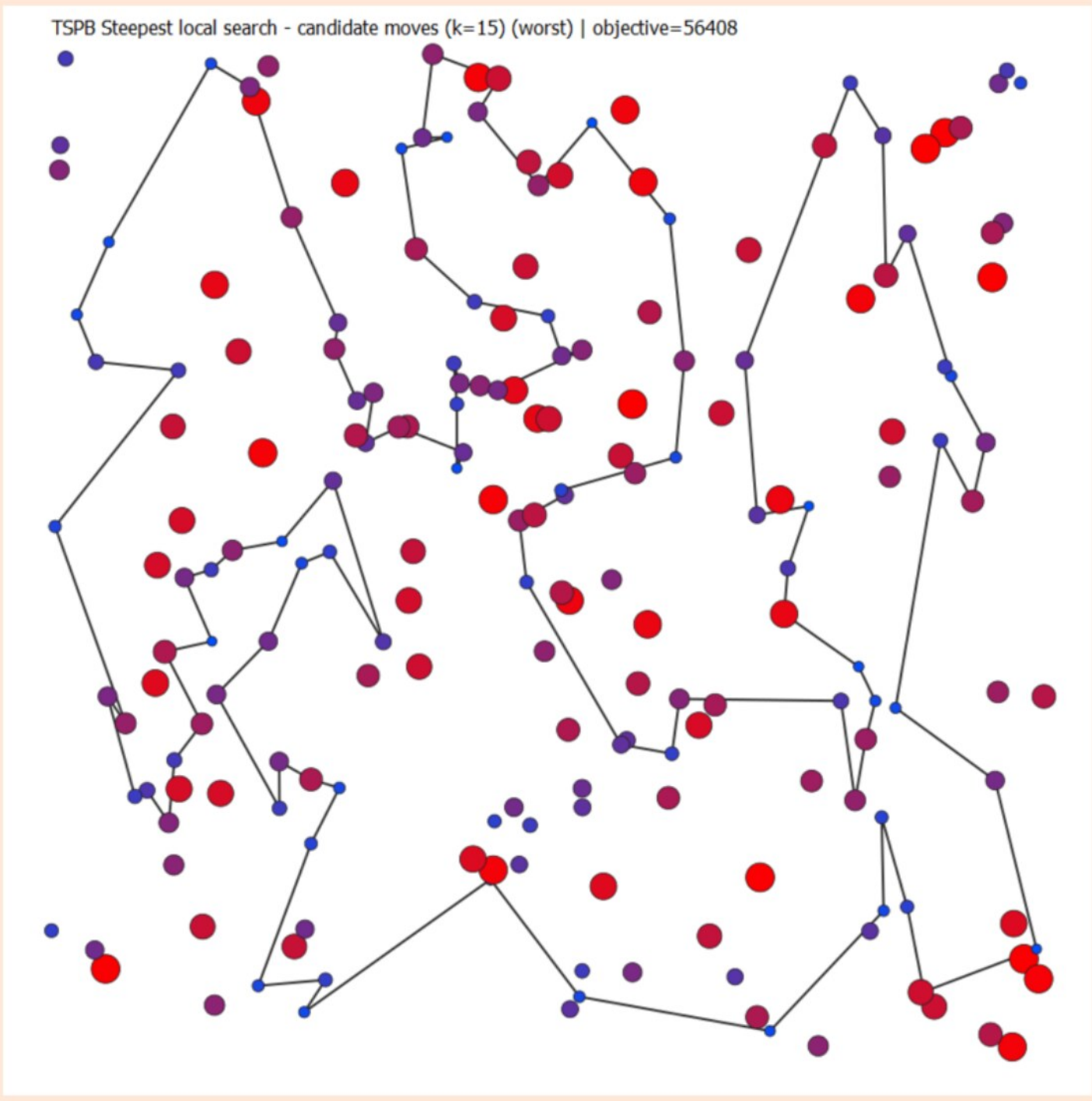
min=51374
max=56520
avg=53573.29

objective 56520

Worst solution nodes:

```
83 79 57 104 76 185 27
135 75 86 72 46 137 117 8
133 172 136 69 160 122
173 184 3 151 98 25 34 24
67 31 66 162 192 54 10
114 196 187 70 81 105 94
127 129 80 90 153 118 110
2 14 20 74 171 87 179 152
181 167 73 59 51 85 60 40
71 45 134 175 132 65 145
28 146 149 158 177 170
125 140 154 47 38 21 30
130 50 116 147 58 164 48
96 198 121 63 120 89 44
83
```

100 selected nodes; closed route



WORST SOLUTION

Candidate moves (k = 15)

min=51171
max=56408
avg=53472.66

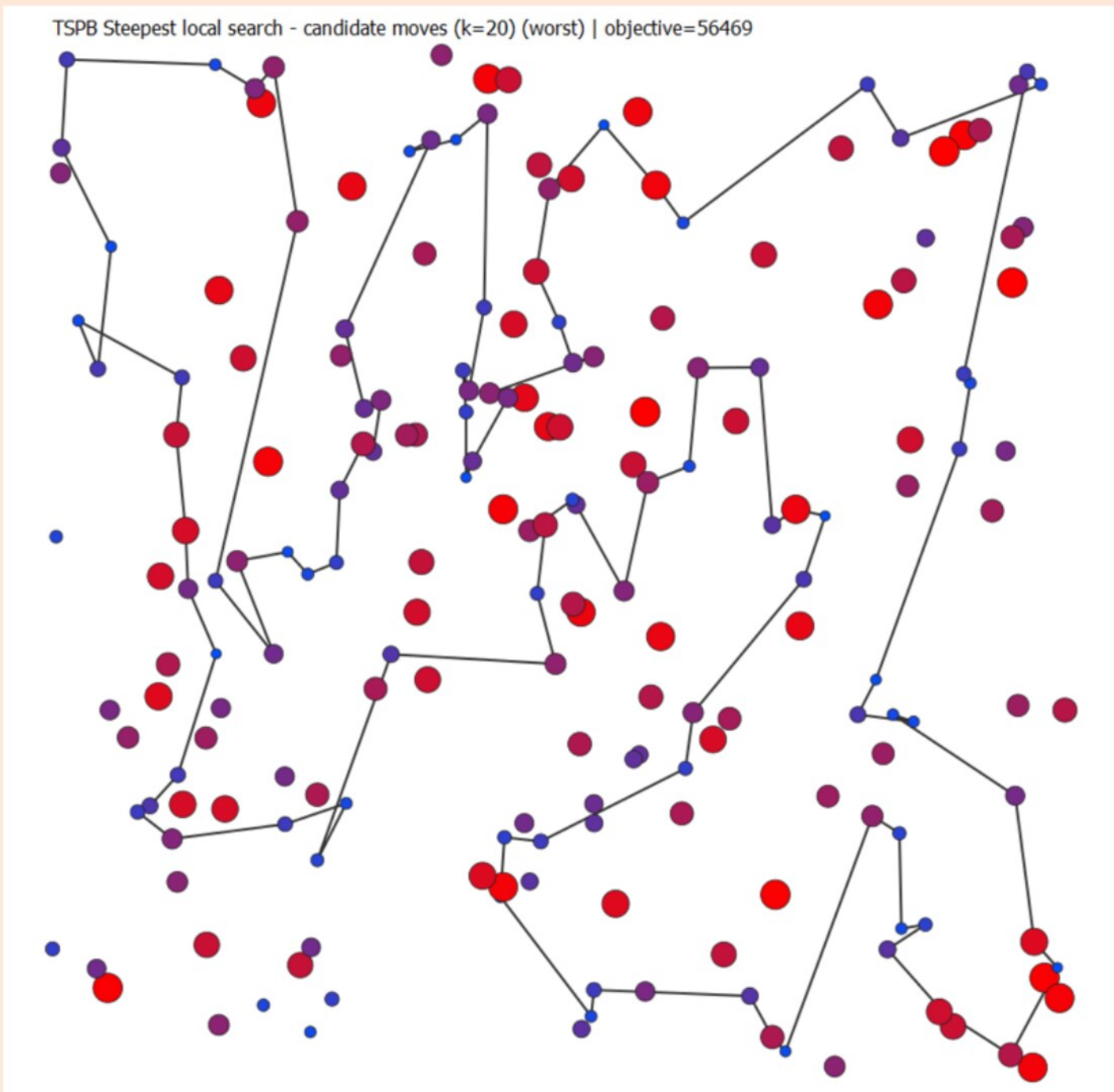
objective 56408

Worst solution nodes:

```
52 182 80 66 67 34 151
122 3 184 76 185 57 89
156 48 164 58 198 96 121
63 120 72 46 86 117 8 136
133 172 69 160 189 21 183
30 191 130 50 147 116 149
28 145 158 65 103 178 132
29 197 6 180 19 170 177
125 26 154 38 64 47 140
97 115 71 155 141 40 60
175 45 134 85 87 171 20
153 14 2 110 118 90 10
143 148 196 114 187 18 70
190 104 162 54 192 106
127 129 52
```

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Candidate moves (k = 20)

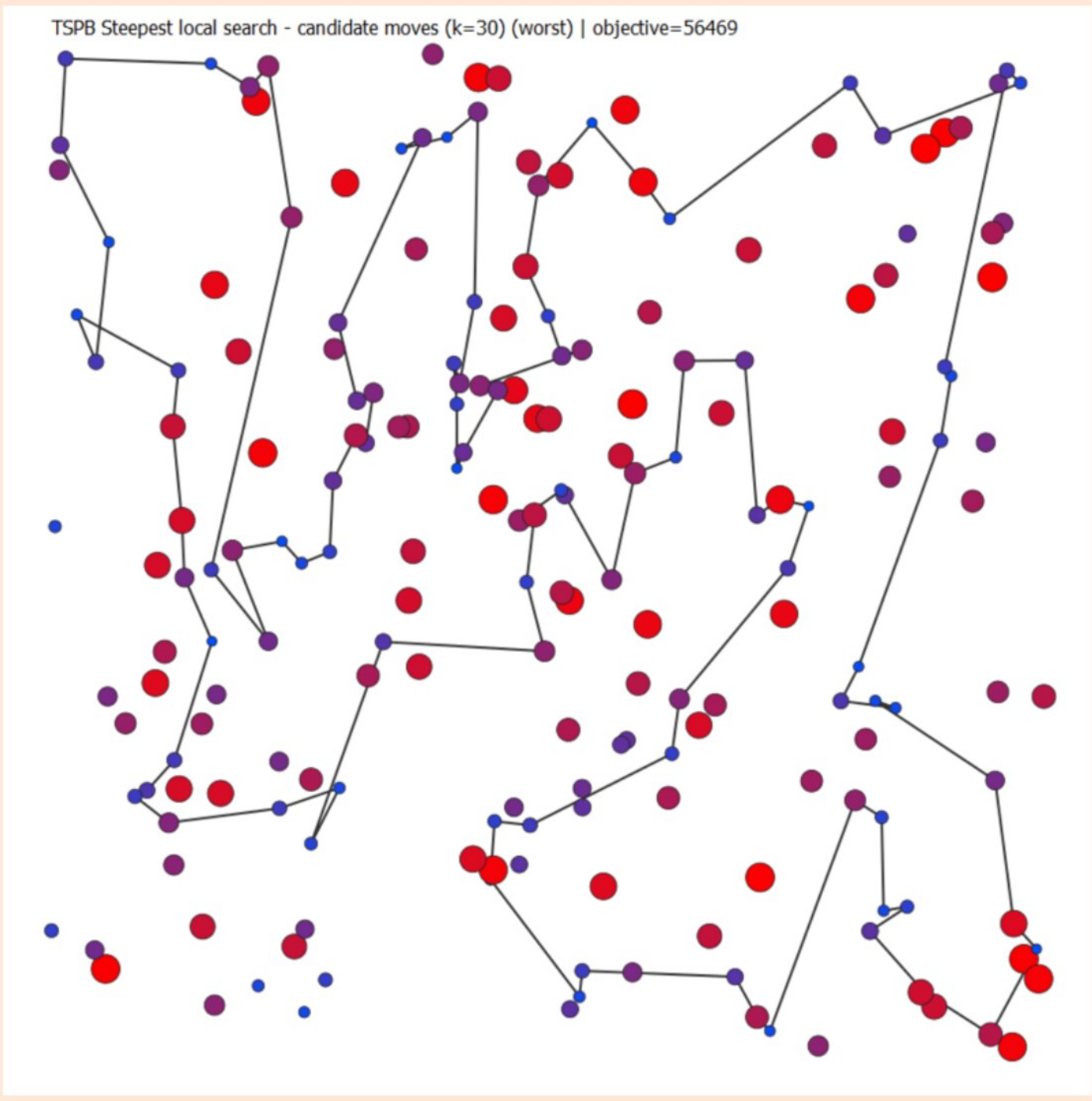
min=51371
max=56469
avg=53461.22

objective 56469

Worst solution nodes:

183 30 130 198 121 96 48
120 136 133 172 69 160
117 8 86 46 72 166 156 89
83 57 190 70 105 94 196
114 10 127 80 90 129 118
68 110 93 2 74 14 20 153
182 171 39 99 179 87 85
59 167 67 66 192 54 159
162 104 185 76 135 27 3
184 122 173 151 98 64 60
40 71 197 132 29 6 170
177 55 138 158 28 145 149
146 37 116 147 42 50 125
97 26 154 140 47 38 176
21 183

100 selected nodes; closed route



WORST SOLUTION

Candidate moves (k = 30)

min=51385
max=56469
avg=53515.31

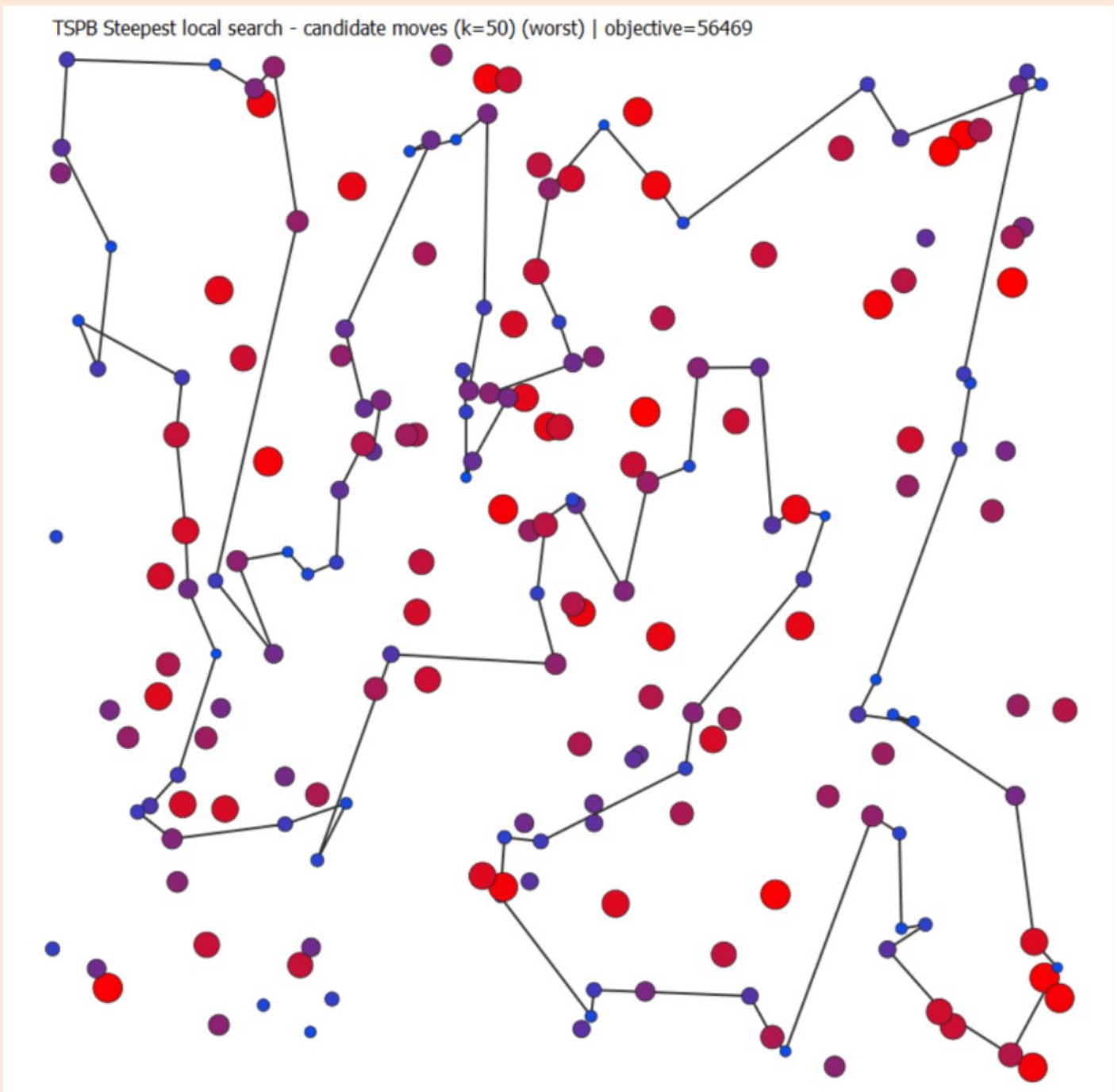
objective 56469

Worst solution nodes:

183 30 130 198 121 96 48
120 136 133 172 69 160
117 8 86 46 72 166 156 89
83 57 190 70 105 94 196
114 10 127 80 90 129 118
68 110 93 2 74 14 20 153
182 171 39 99 179 87 85
59 167 67 66 192 54 159
162 104 185 76 135 27 3
184 122 173 151 98 64 60
40 71 197 132 29 6 170
177 55 138 158 28 145 149
146 37 116 147 42 50 125
97 26 154 140 47 38 176
21 183

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Candidate moves (k = 50)

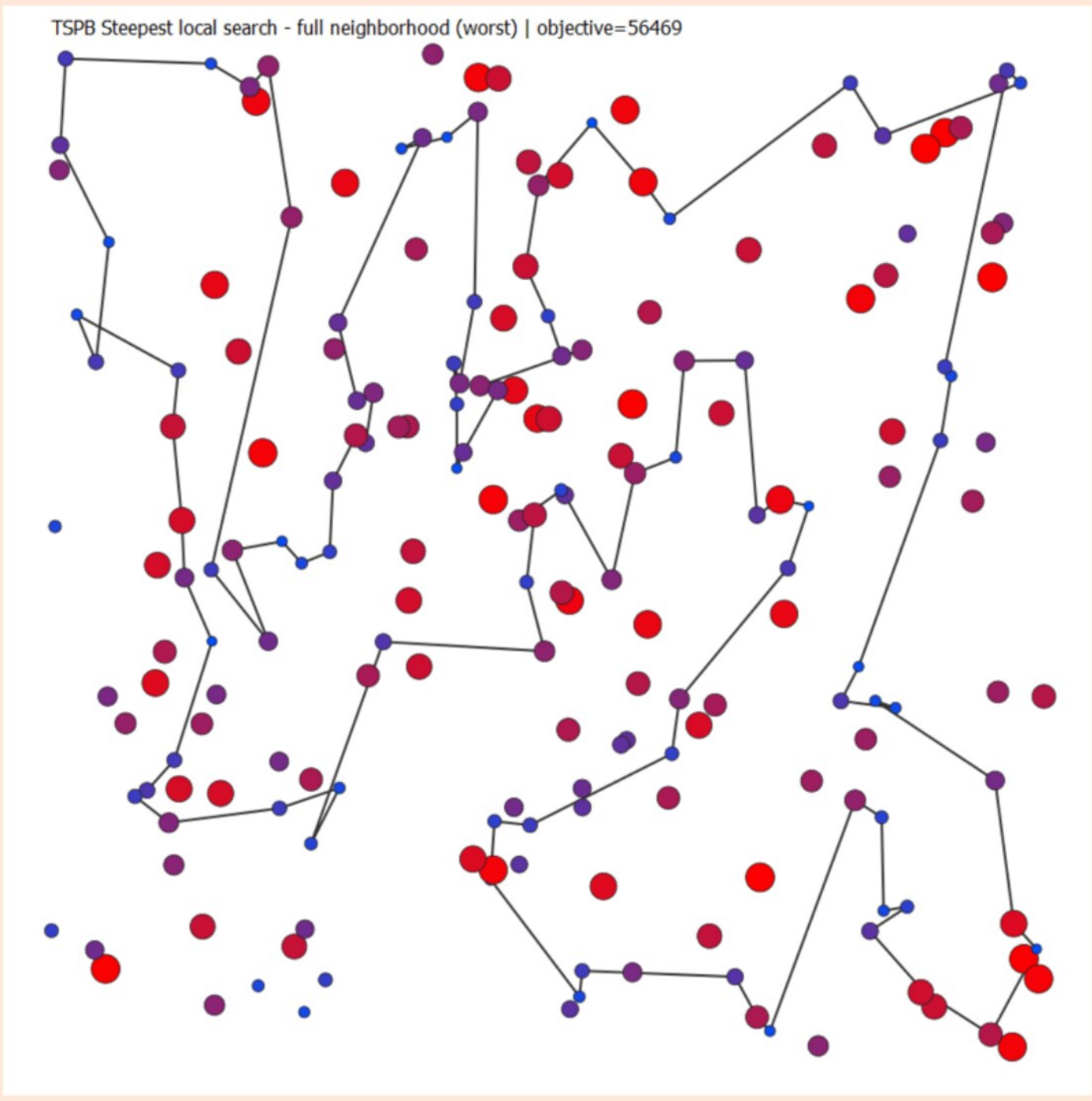
min=51385
max=56469
avg=53506.79

objective 56469

Worst solution nodes:

183 30 130 198 121 96 48
120 136 133 172 69 160
117 8 86 46 72 166 156 89
83 57 190 70 105 94 196
114 10 127 80 90 129 118
68 110 93 2 74 14 20 153
182 171 39 99 179 87 85
59 167 67 66 192 54 159
162 104 185 76 135 27 3
184 122 173 151 98 64 60
40 71 197 132 29 6 170
177 55 138 158 28 145 149
146 37 116 147 42 50 125
97 26 154 140 47 38 176
21 183

100 selected nodes; closed route



WORST SOLUTION

Full neighborhood (k = infinity)

min=51385
max=56469
avg=53505.13

objective 56469

Worst solution nodes:

183 30 130 198 121 96 48
120 136 133 172 69 160
117 8 86 46 72 166 156 89
83 57 190 70 105 94 196
114 10 127 80 90 129 118
68 110 93 2 74 14 20 153
182 171 39 99 179 87 85
59 167 67 66 192 54 159
162 104 185 76 135 27 3
184 122 173 151 98 64 60
40 71 197 132 29 6 170
177 55 138 158 28 145 149
146 37 116 147 42 50 125
97 26 154 140 47 38 176
21 183

100 selected nodes; closed route